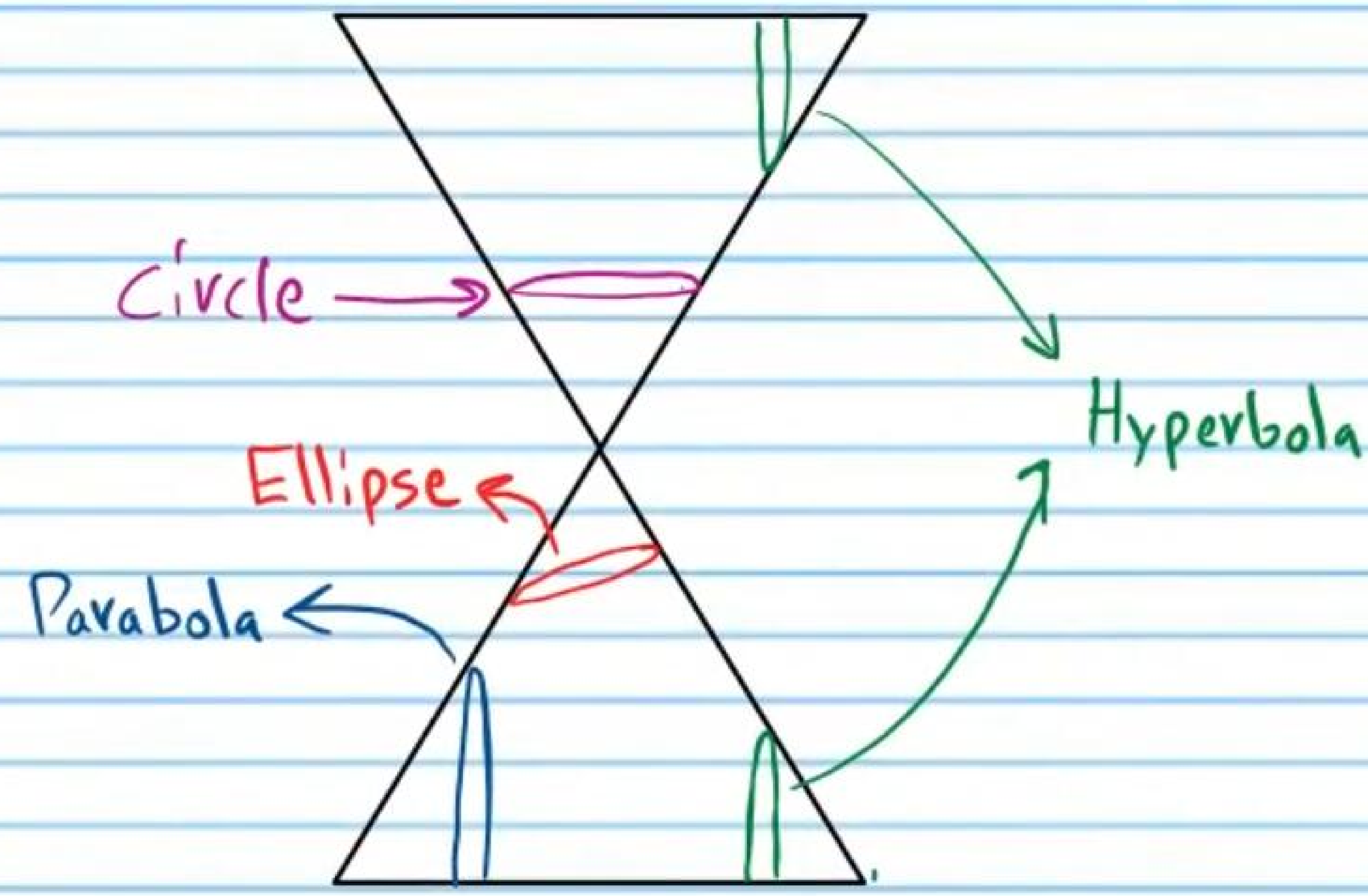
Name
Date
Course

The Conics (Part I)

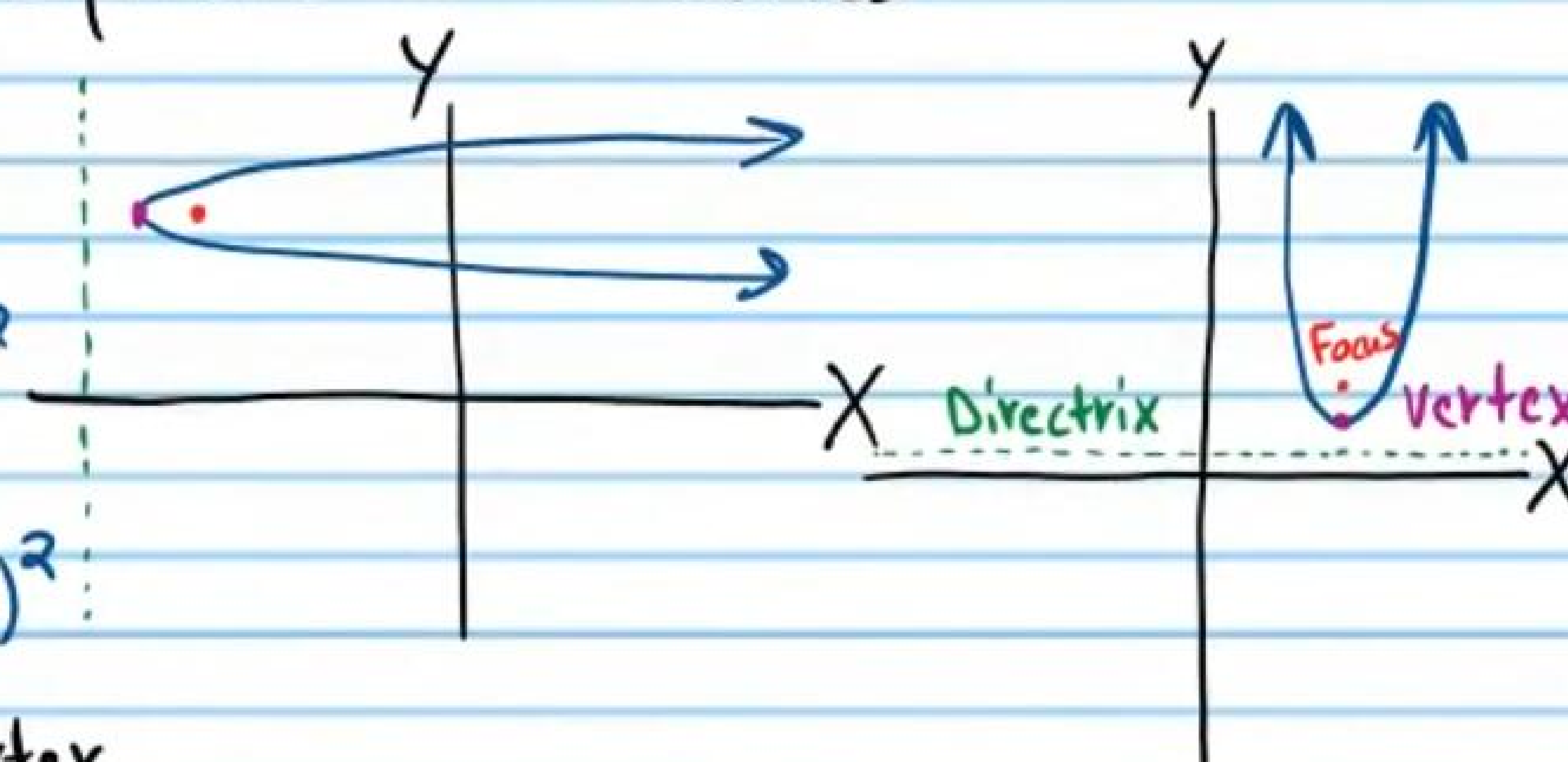


Equations of the Conics

Parabola

Vertical
$y-k=a(x-h)^2$

Horizontal
$x-h=a(y-k)^2$



(h,k) is the vertex.

$a=\frac{1}{4p}$,

p is the distance from the vertex to the focus or directrix.

Other Forms

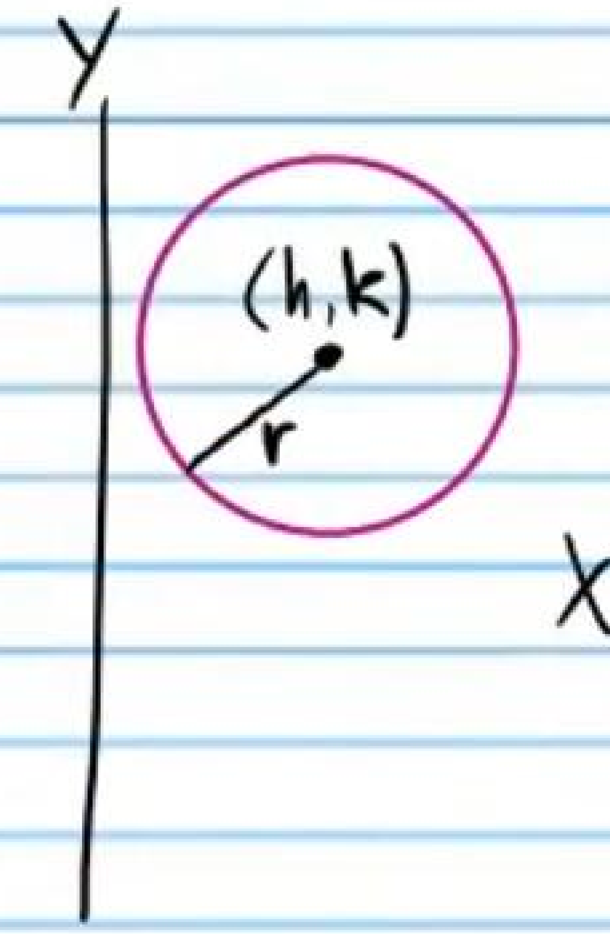
Standard $y=ax^2+bx+c$

Intercept $y=a(x-m)(x-n)$

Circle

$(x-h)^2+(y-k)^2=r^2$

(h,k) is the center.
r is the radius.

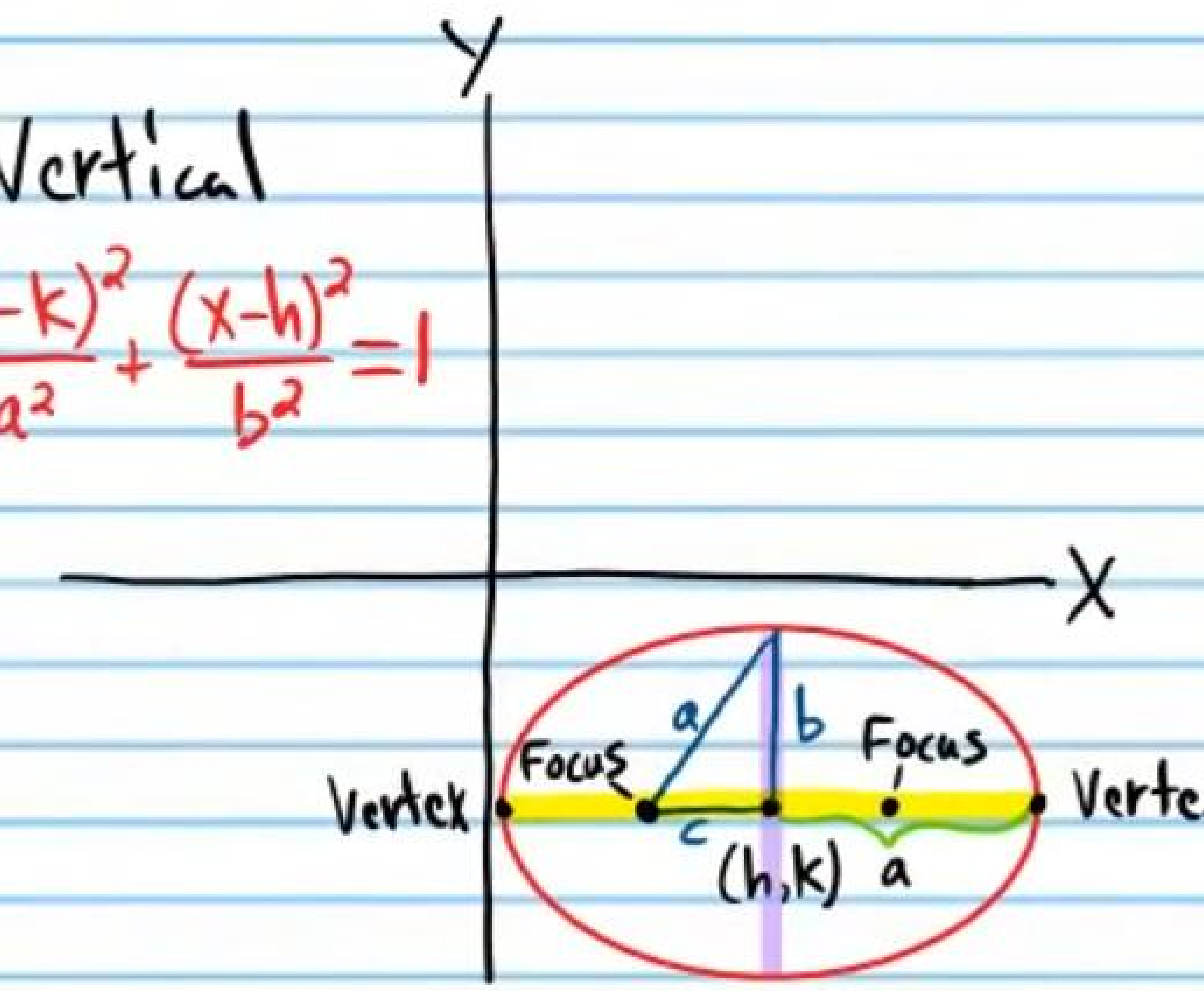


Ellipse

Horizontal
$$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$$

Vertical
$$\frac{(y-k)^2}{a^2}+\frac{(x-h)^2}{b^2}=1$$

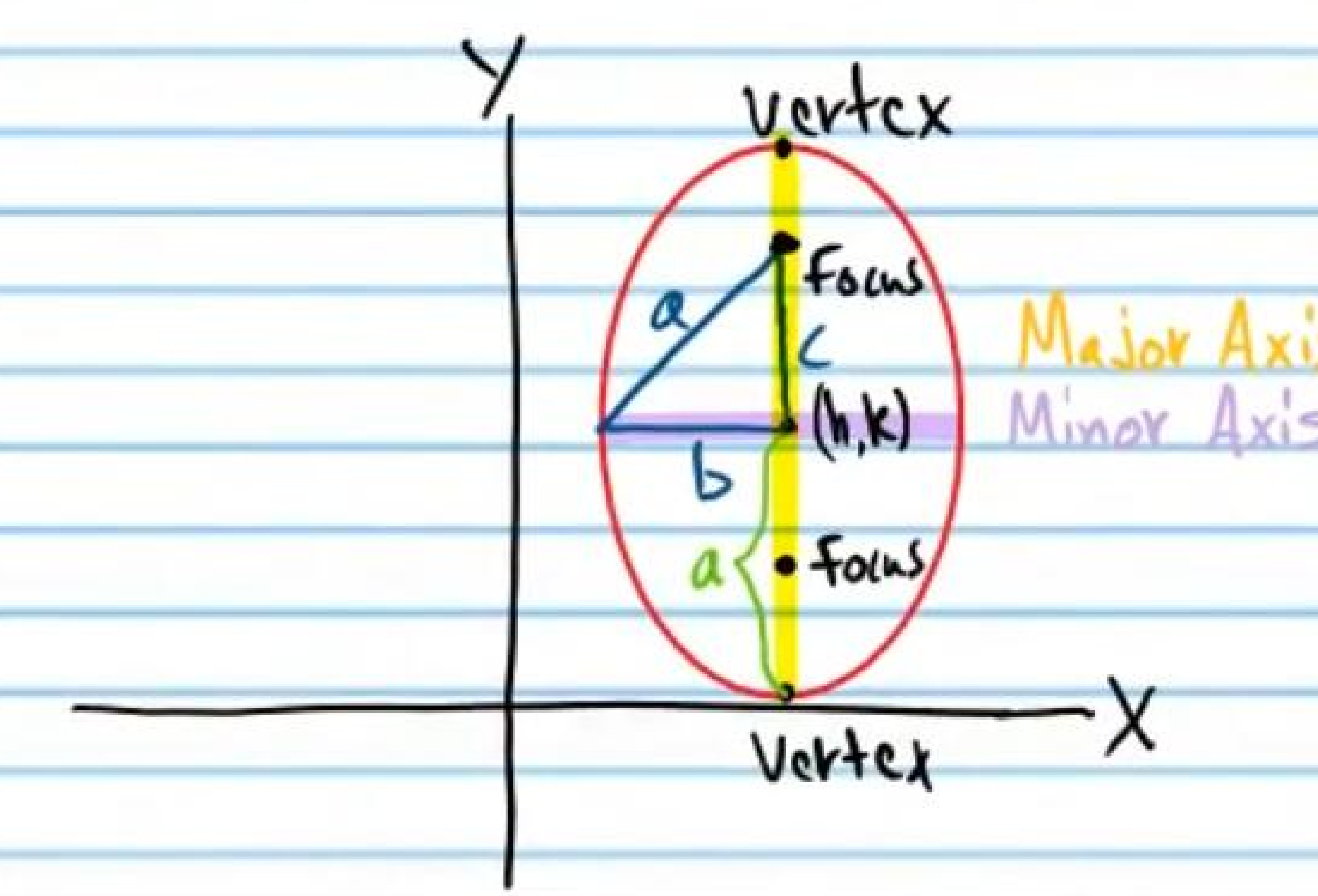
Vertical



(h,k) is the center.

$c^2+b^2=a^2$

c is the distance along the major axis from the center to the foci. a is the distance from the center to the vertices. 2a is the length of the major axis. 2b is the length of the minor axis.



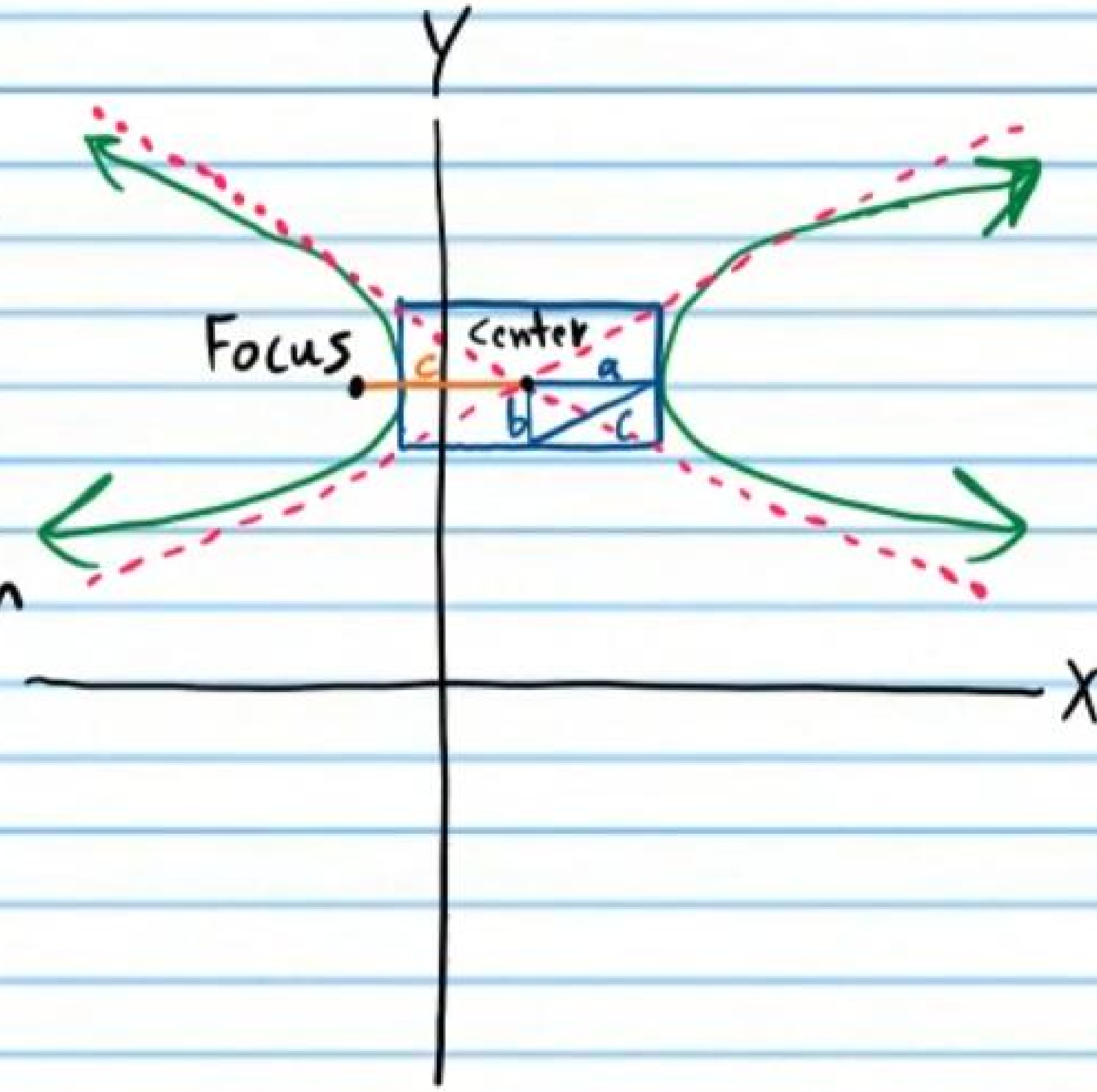
Hyperbola

Opens Right and Left

$$\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$$

Opens Up and Down

$$\frac{(y-k)^2}{a^2}-\frac{(x-h)^2}{b^2}=1$$



(h,k) is the center.

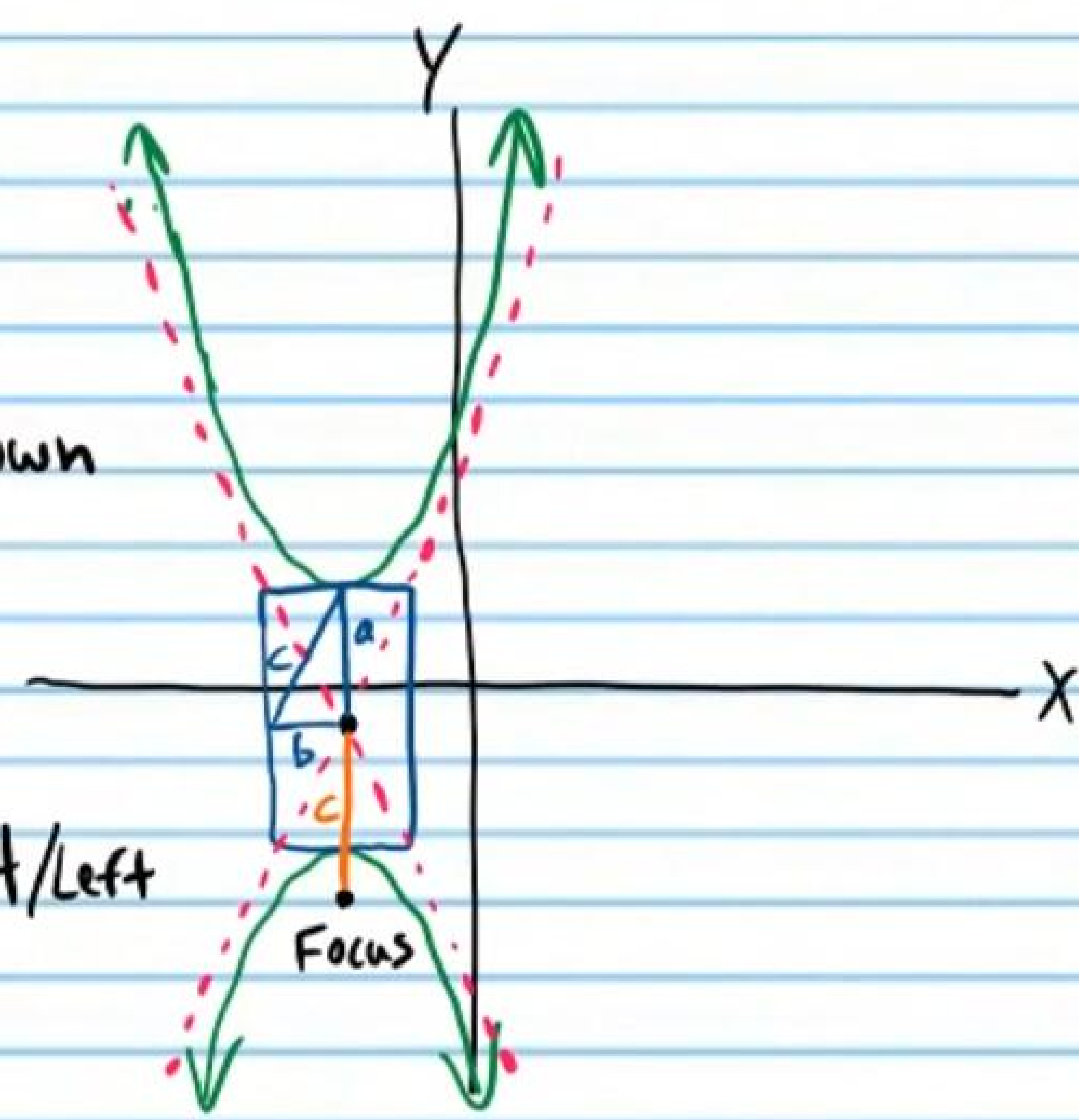
$a^2+b^2=c^2$

c is the distance from the center to the foci. a is the distance from the center to the vertices.

Asymptotes Opening Up/Down

$y-k=\frac{a}{b}(x-h)$

$y-k=-\frac{a}{b}(x-h)$



Asymptotes Opening Right/Left

$x-h=\frac{b}{a}(y-k)$

$x-h=-\frac{b}{a}(y-k)$

$(x-h)^2+(y-k)^2=r^2$

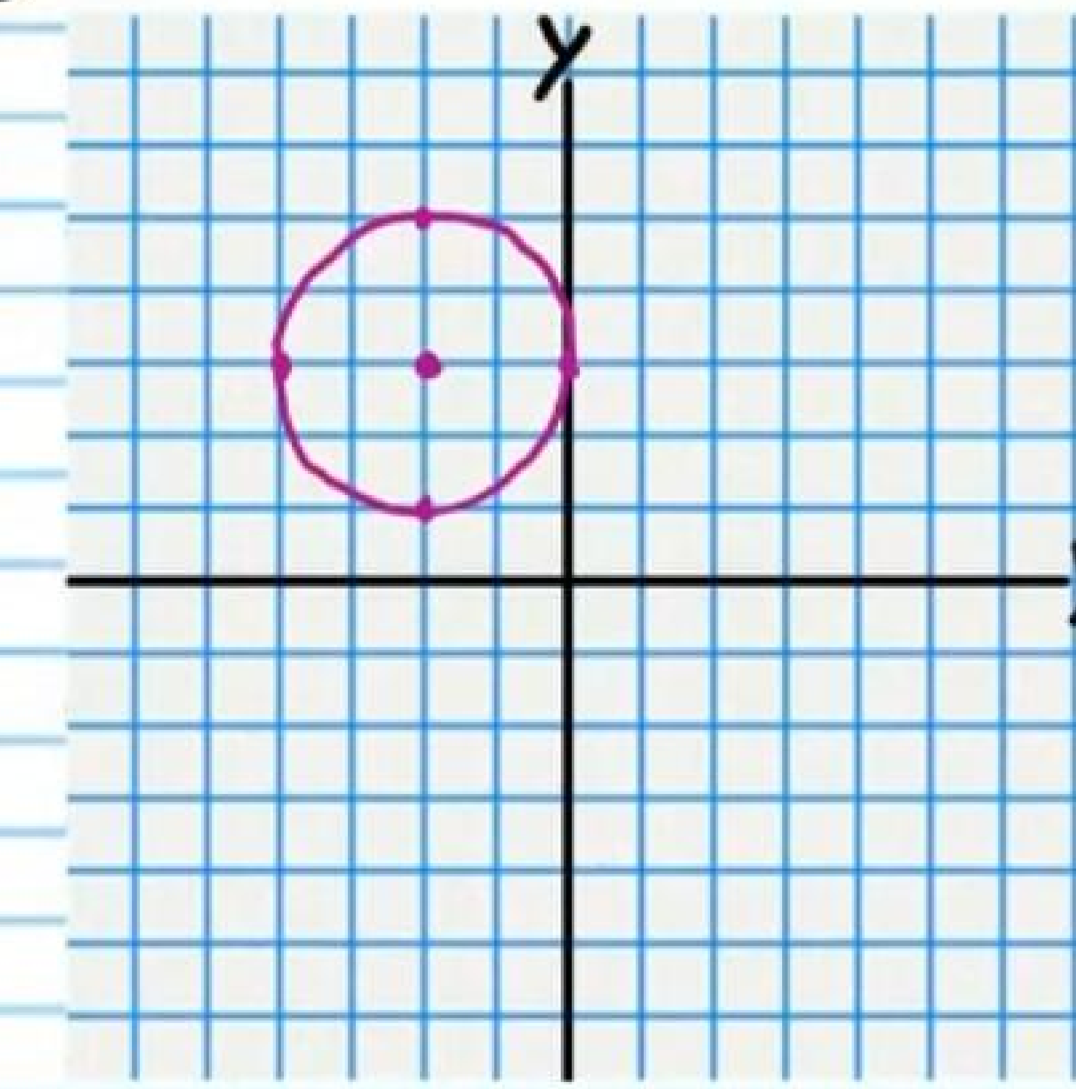
Circles

Graph the following equations. Write the coordinates of the center and the length of the radius.

① $(x+2)^2+(y-3)^2=4$

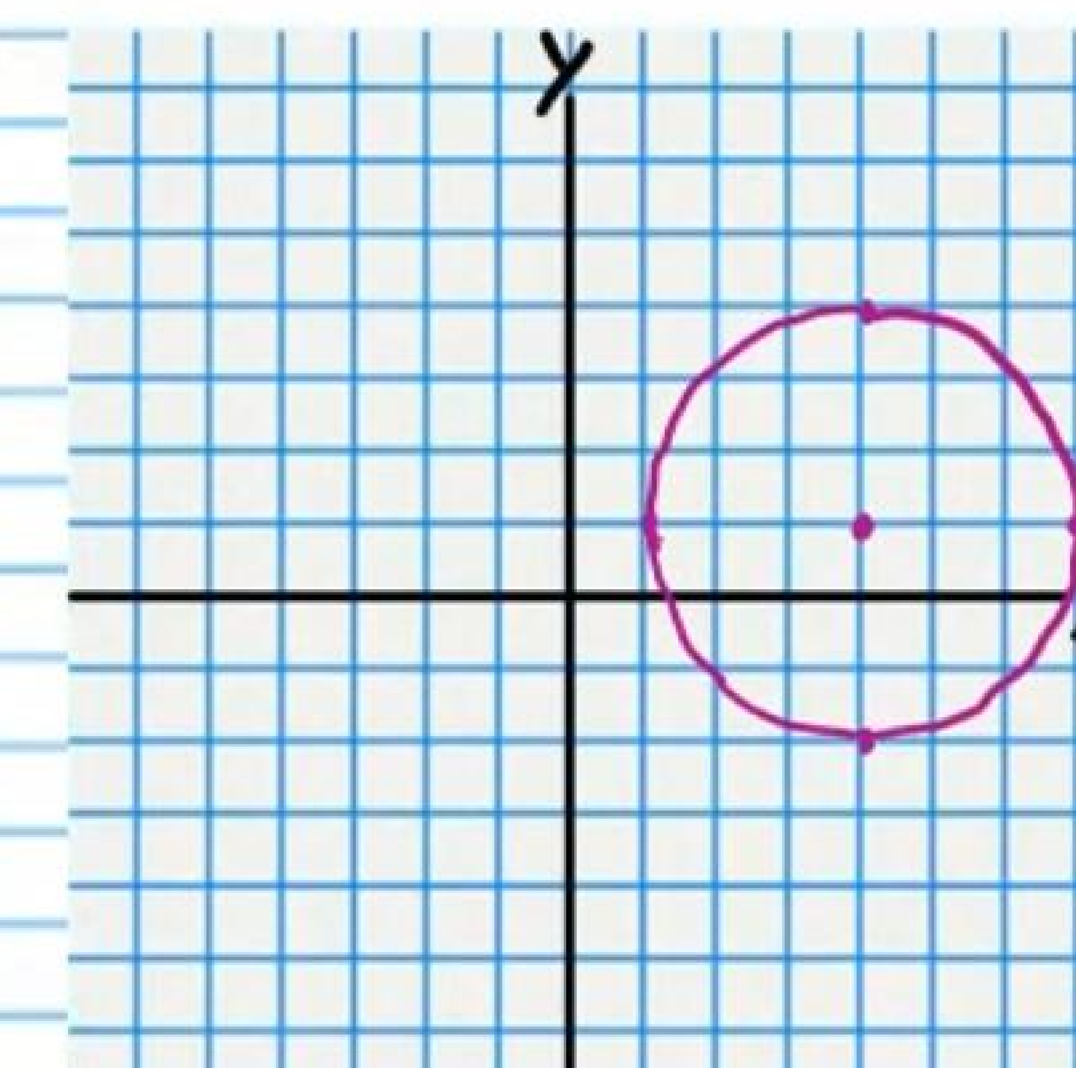
$(x+2)^2+(y-3)^2=2^2$

$\boxed{C(-2,3) \quad r=2}$



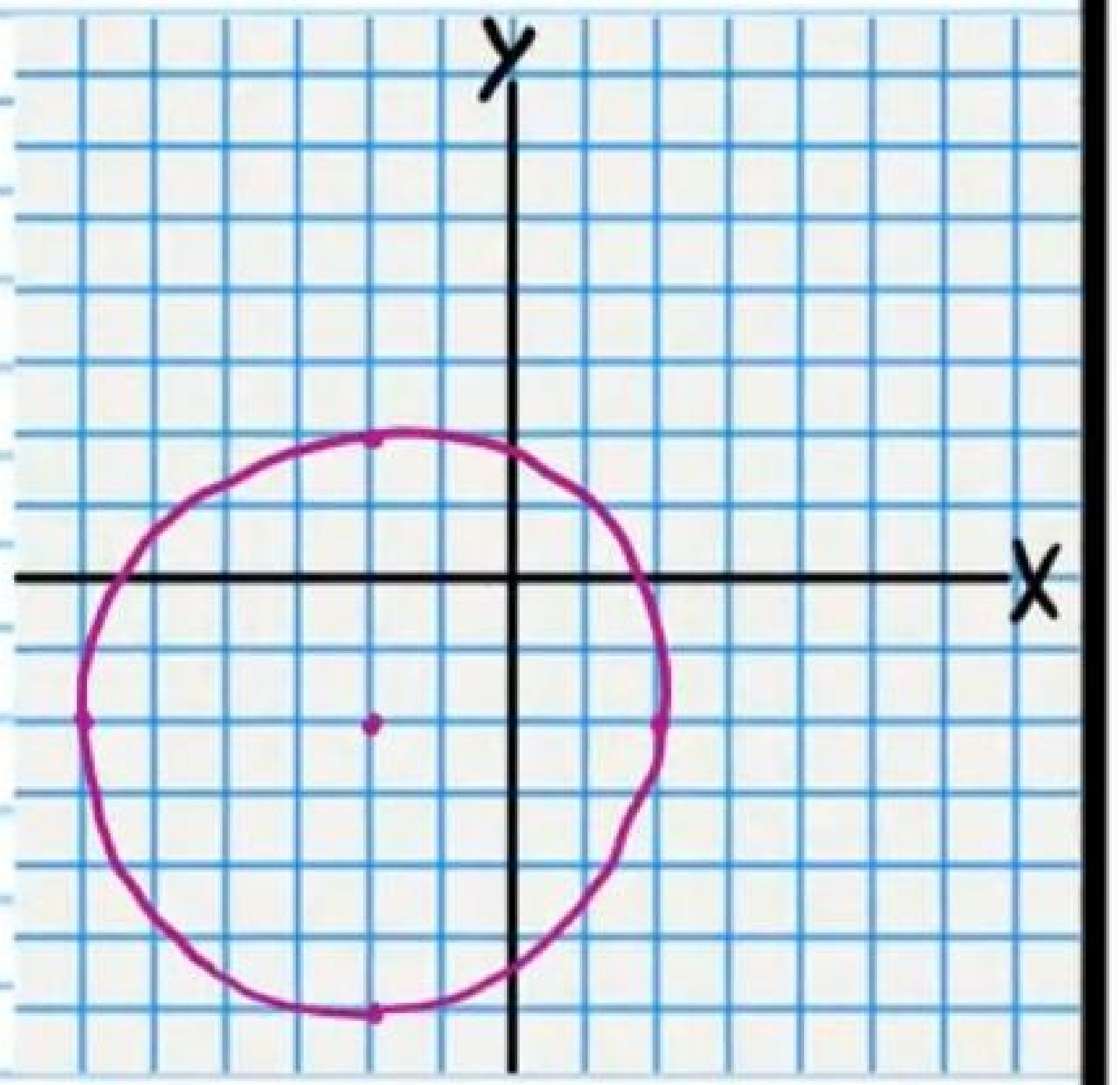
② $9=(x-4)^2+(y-1)^2$

$\boxed{C(4,1) \quad r=3}$



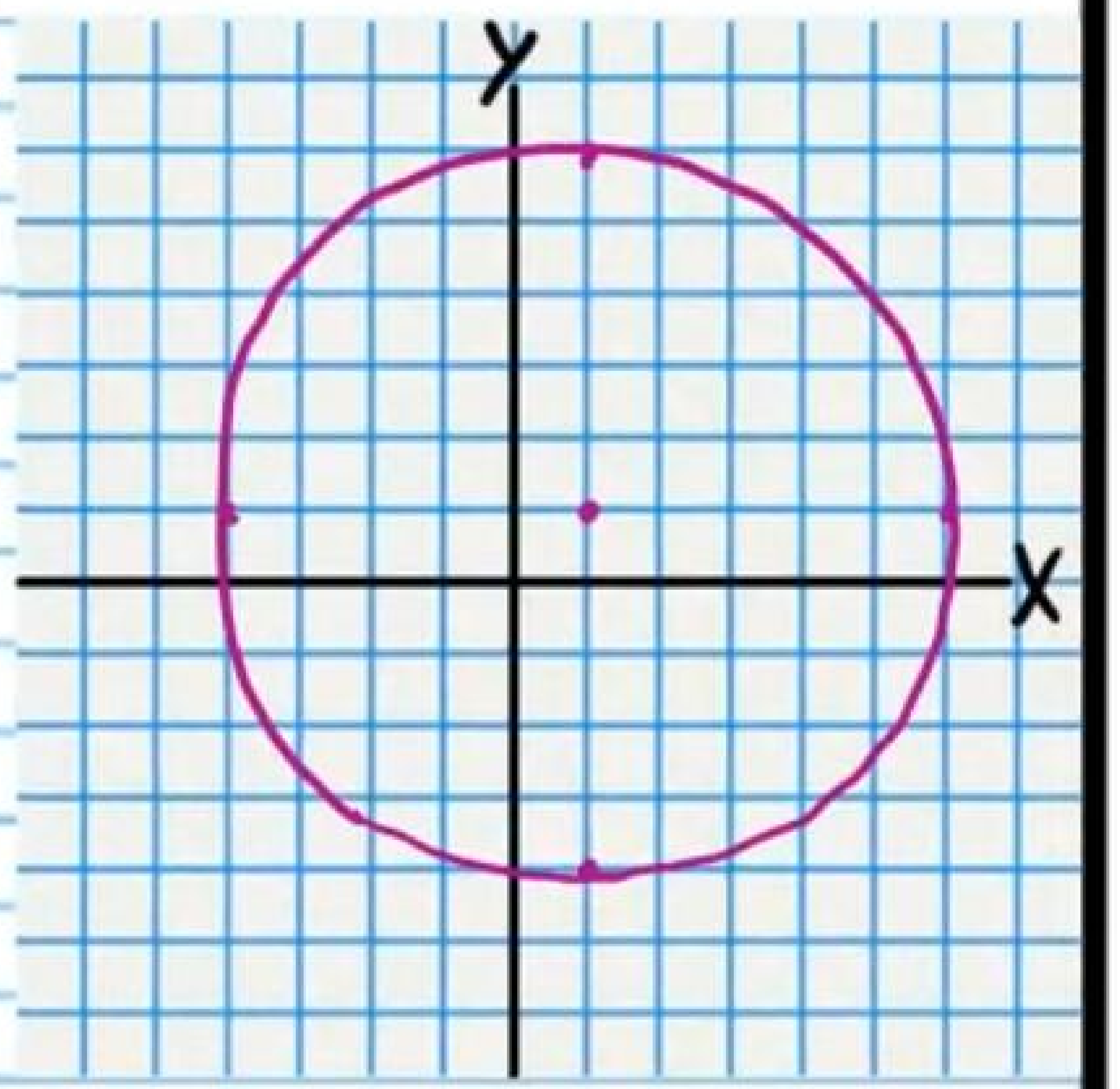
③ $(x+2)^2+(y+2)^2=16$

$\boxed{C(-2,-2) \quad r=4}$



④ $25=(x-1)^2+(y-1)^2$

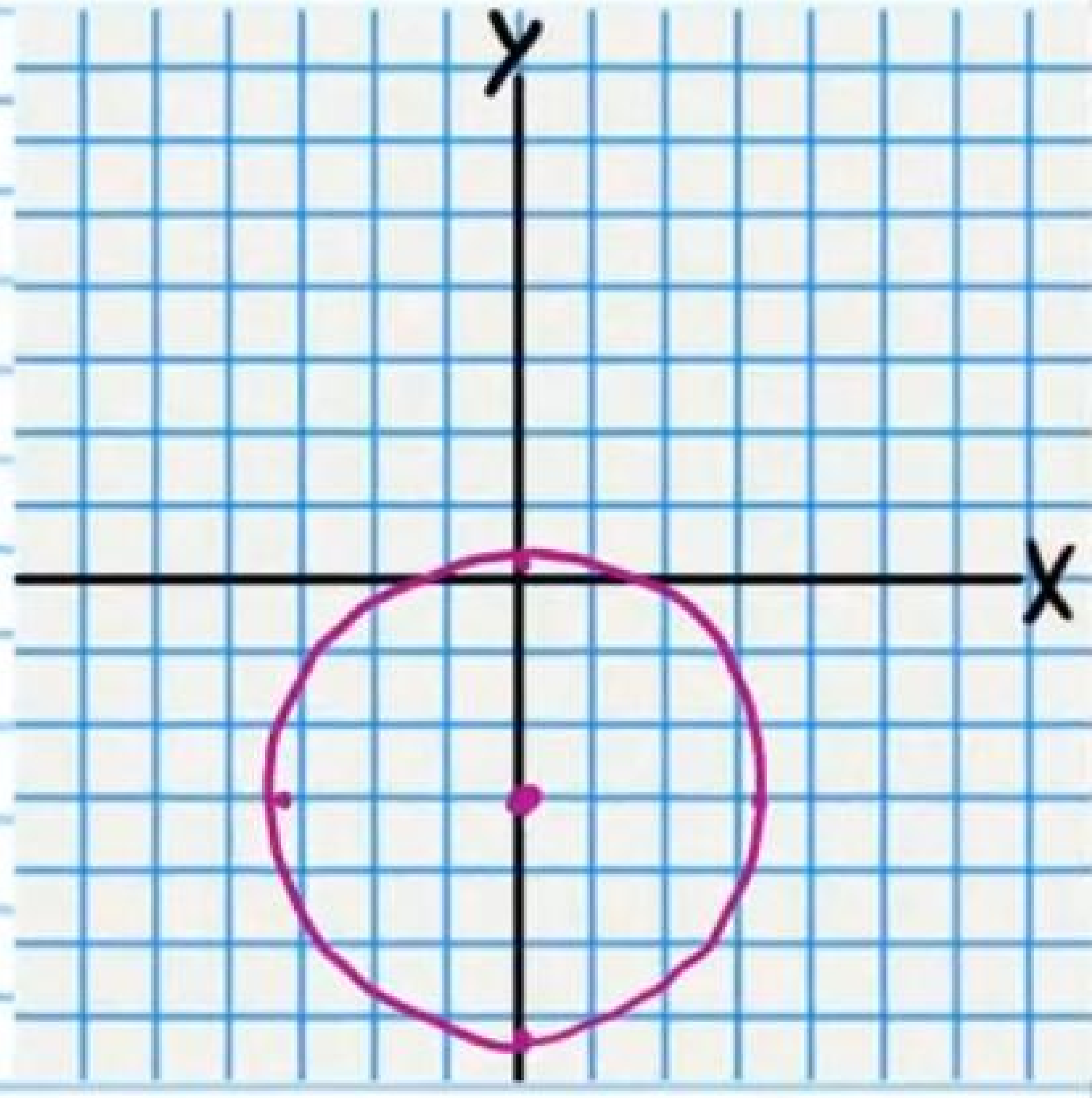
$\boxed{C(1,1) \quad r=5}$



⑤ $x^2+(y+3)^2=11$

$(x-0)^2+(y+3)^2=(\sqrt{11})^2$

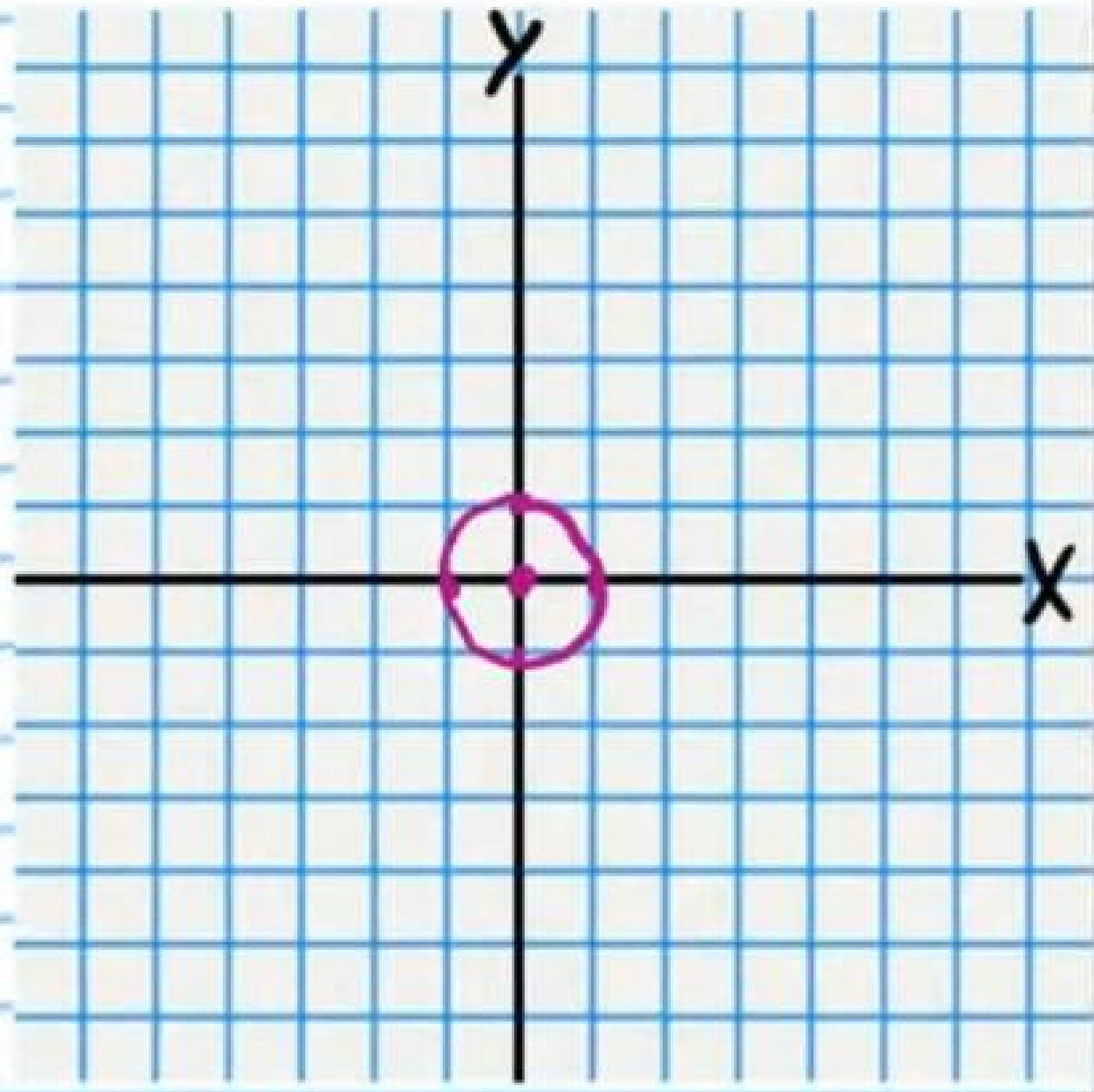
$\boxed{C(0,-3) \quad r=\sqrt{11}}$



⑥ $x^2+y^2=1$

$(x-0)^2+(y-0)^2=1^2$

$\boxed{C(0,0) \quad r=1}$



$y-k=a(x-h)^2$ $\qquad$ $x-h=a(y-k)^2$

Parabolas

Graph the following equations. Write the coordinates of the focus and the equation of the directrix.

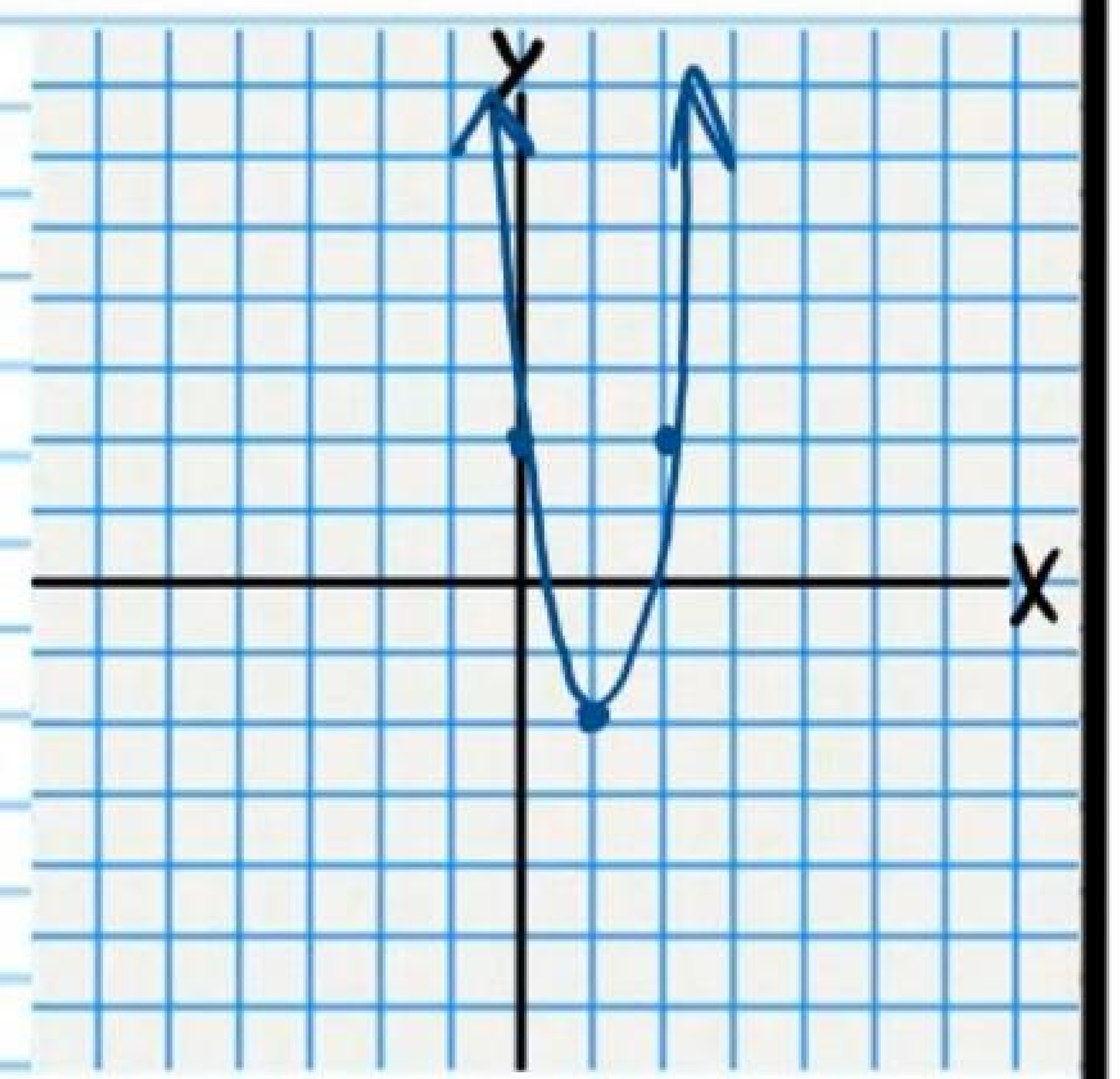
⑦ $y+2=4(x-1)^2$

$P=\frac{1}{4a}=\frac{1}{4(4)}=\frac{1}{16}$

$F(1,-2+\frac{1}{16})=F(1,-1\frac{15}{16})$

$y=-2-\frac{1}{16}$

$\boxed{y=-2\frac{1}{16}}$



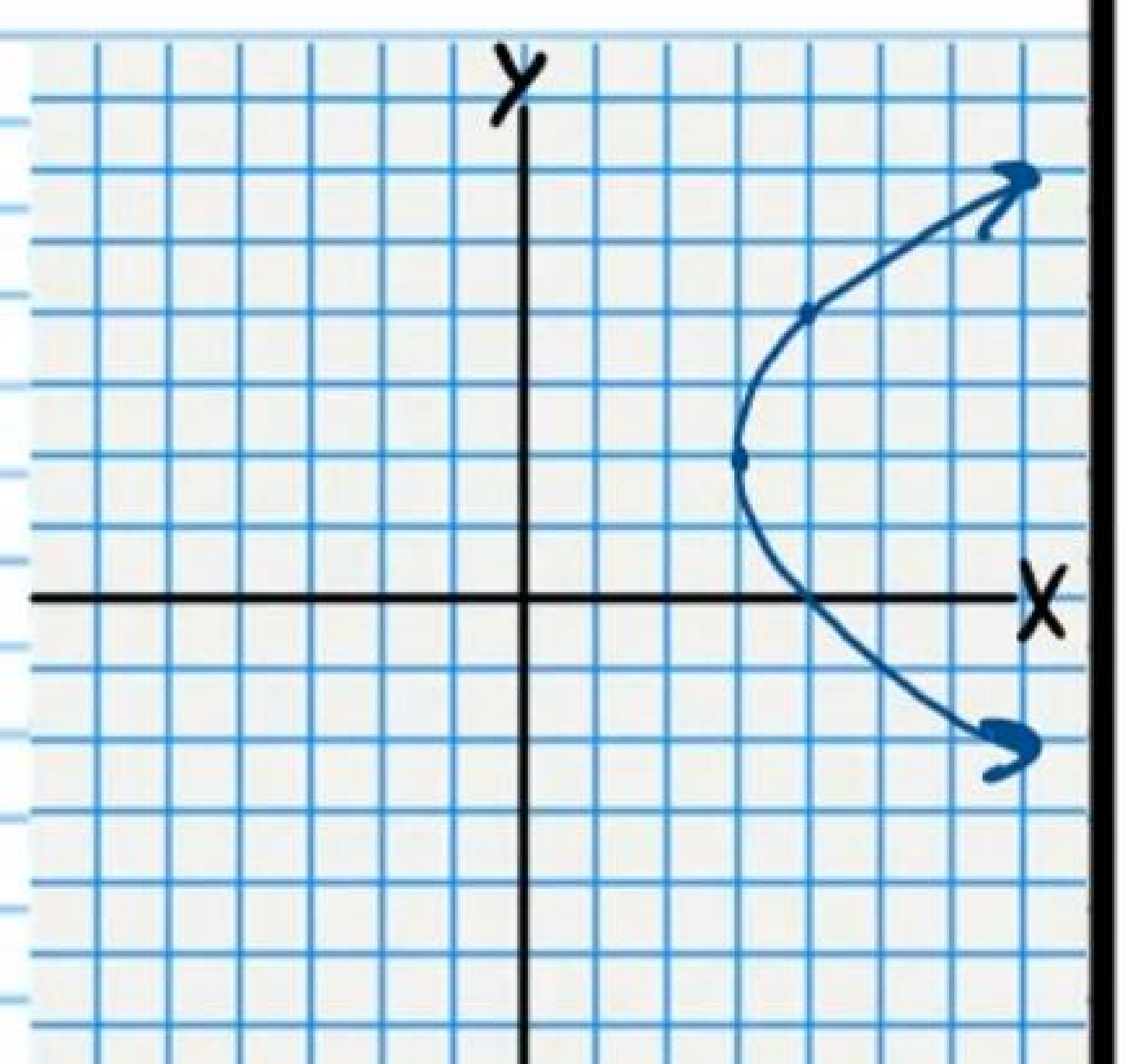
⑧ $\frac{1}{4}(y-2)^2=x-3$

$P=\frac{1}{4a}=\frac{1}{4(\frac{1}{4})}=\frac{1}{1}=1$

$F(3+1,2)=\boxed{F(4,2)}$

$x=3-1$

$\boxed{x=2}$



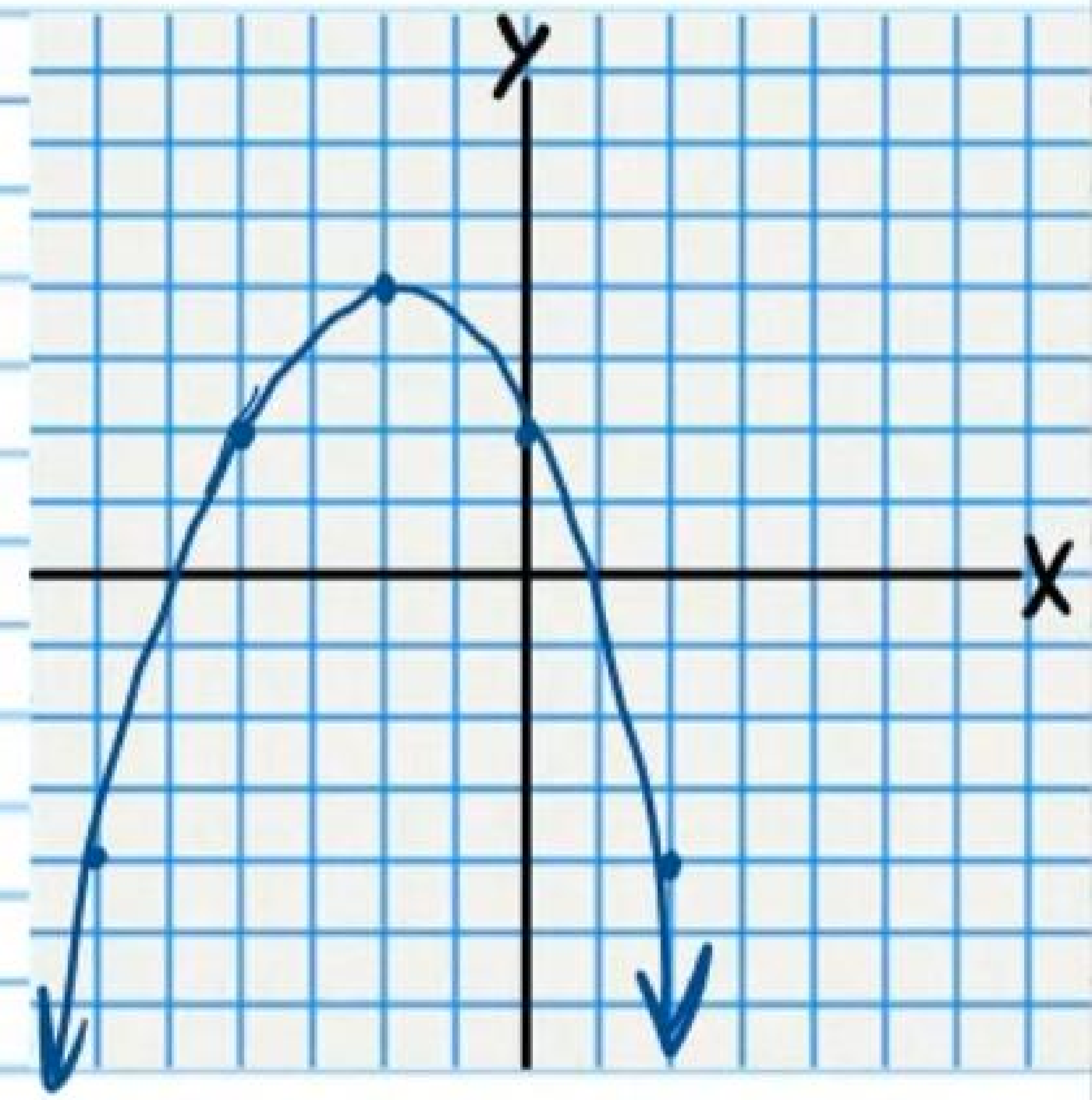
⑨ $-\frac{1}{2}(x+2)^2=y-4$

$P=\frac{1}{4a}=\frac{1}{4(-\frac{1}{2})}=-\frac{1}{2}$

$F(-2,4-\frac{1}{2})=\boxed{F(-2,3\frac{1}{2})}$

$y=4+\frac{1}{2}$

$\boxed{y=4\frac{1}{2}}$



⑩ $x=-3(y+1)^2$

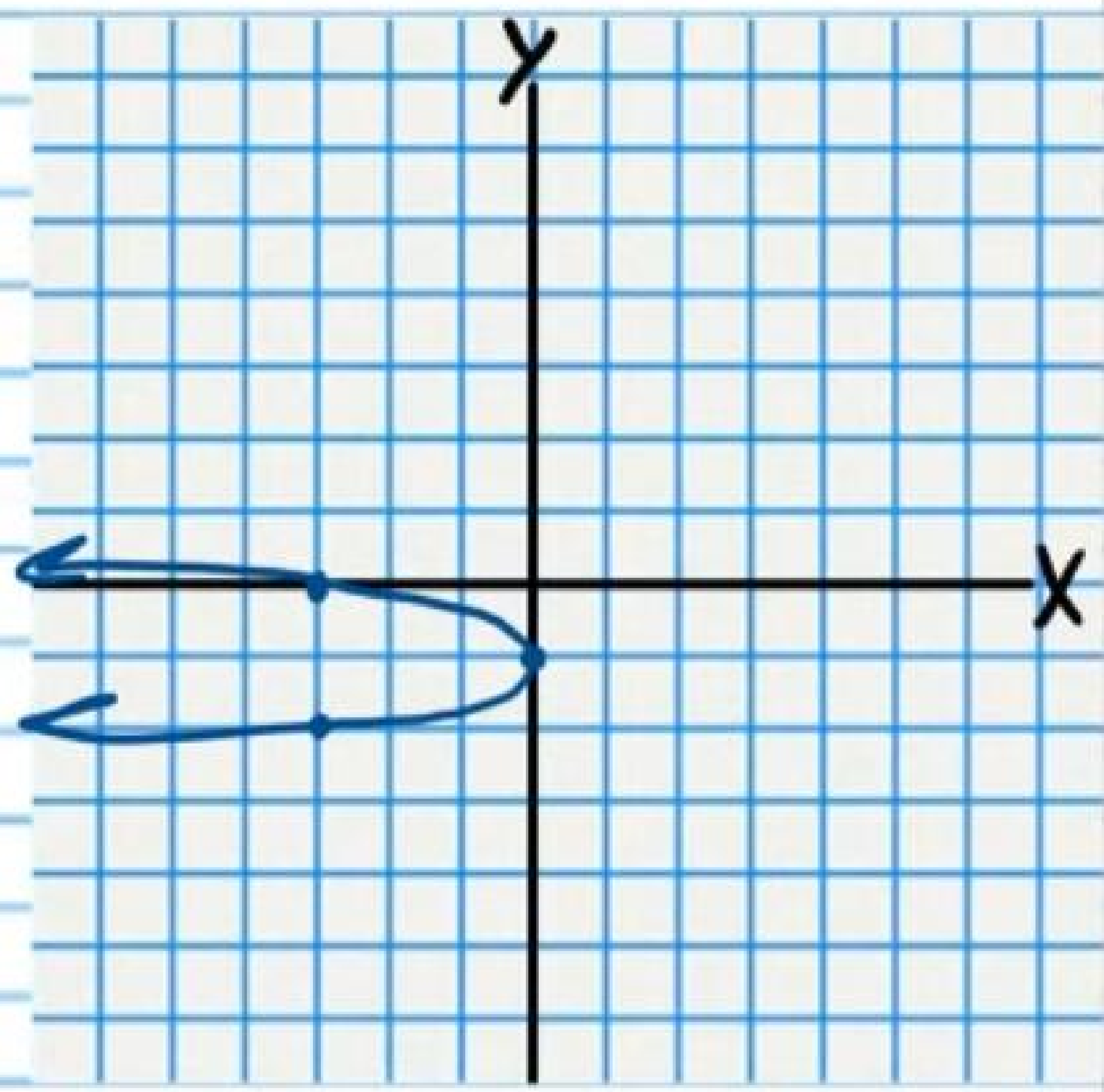
$x-0=-3(y+1)^2$

$P=\frac{1}{4a}=\frac{1}{4(-3)}=-\frac{1}{12}$

$F(0-\frac{1}{12},-1)=\boxed{F(-\frac{1}{12},-1)}$

$x=0+\frac{1}{12}$

$\boxed{x=\frac{1}{12}}$



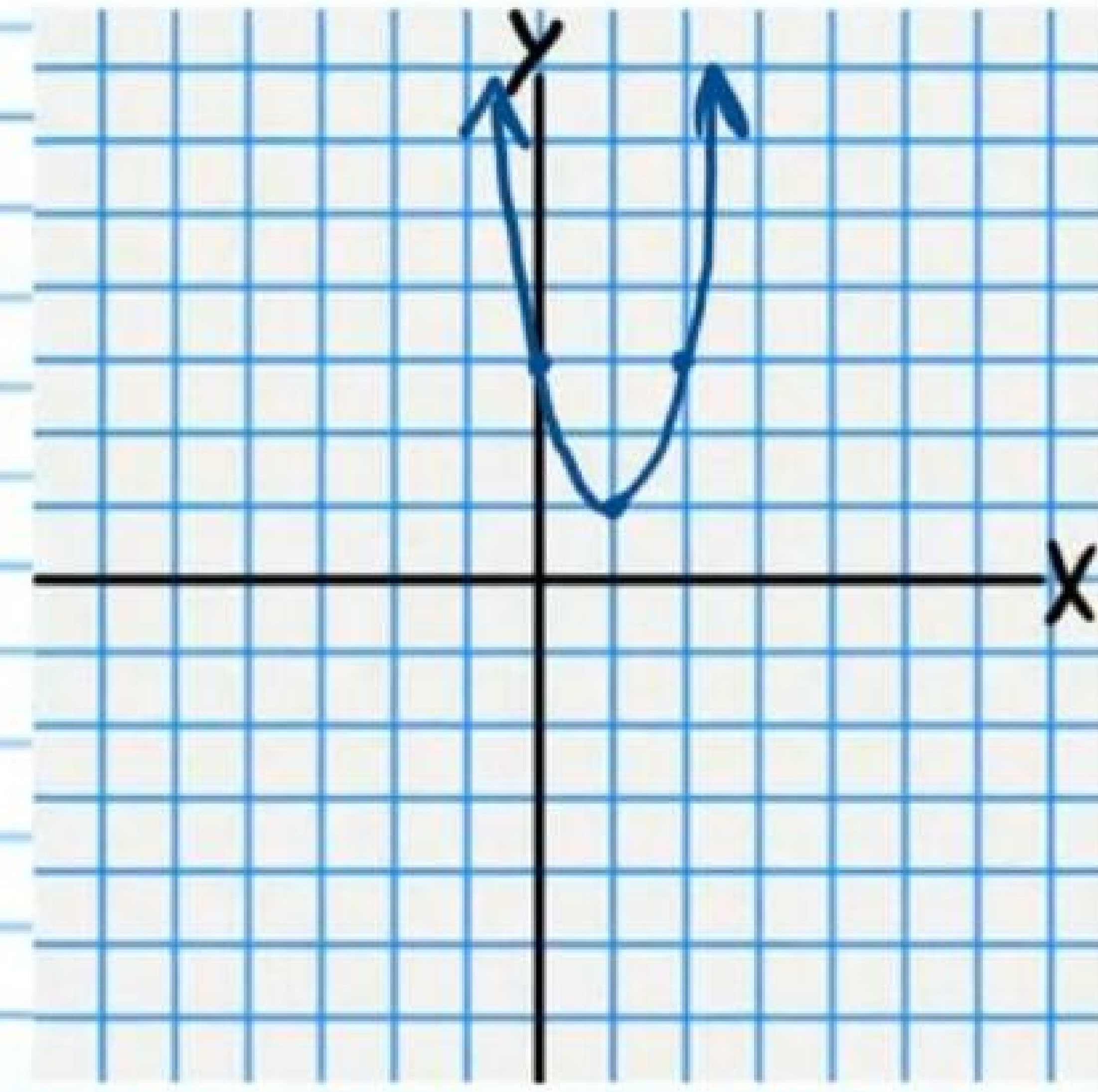
(11) $y-1=2(x-1)^2$

$$P=\frac{1}{4a}=\frac{1}{4(2)}=\frac{1}{8}$$

$$F\left(1,1+\tfrac{1}{8}\right)=\boxed{F\left(1,1\tfrac{1}{8}\right)}$$

$$y=1-\frac{1}{8}$$

$$\boxed{y=\frac{7}{8}}$$



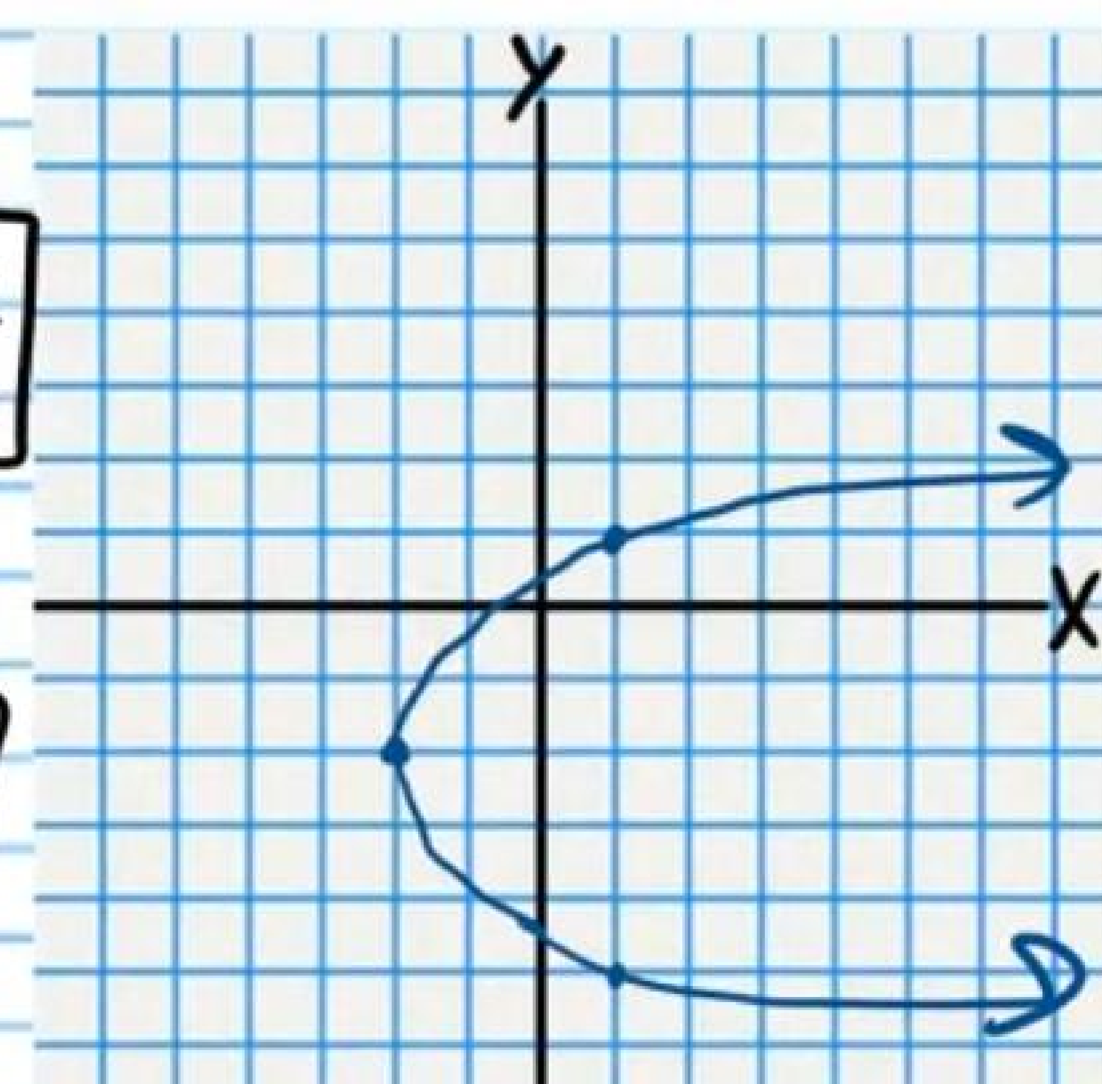
(12) $\frac{1}{3}(y+2)^2=x+2$

$$P=\frac{1}{4a}=\frac{1}{4\left(\frac{1}{3}\right)}=\frac{1}{\frac{4}{3}}=\boxed{\frac{3}{4}}$$

$$F\left(-2+\tfrac{3}{4},-2\right)=\boxed{F\left(-1\tfrac{1}{4},-2\right)}$$

$$x=-2-\frac{3}{4}$$

$$\boxed{x=-2\frac{3}{4}}$$



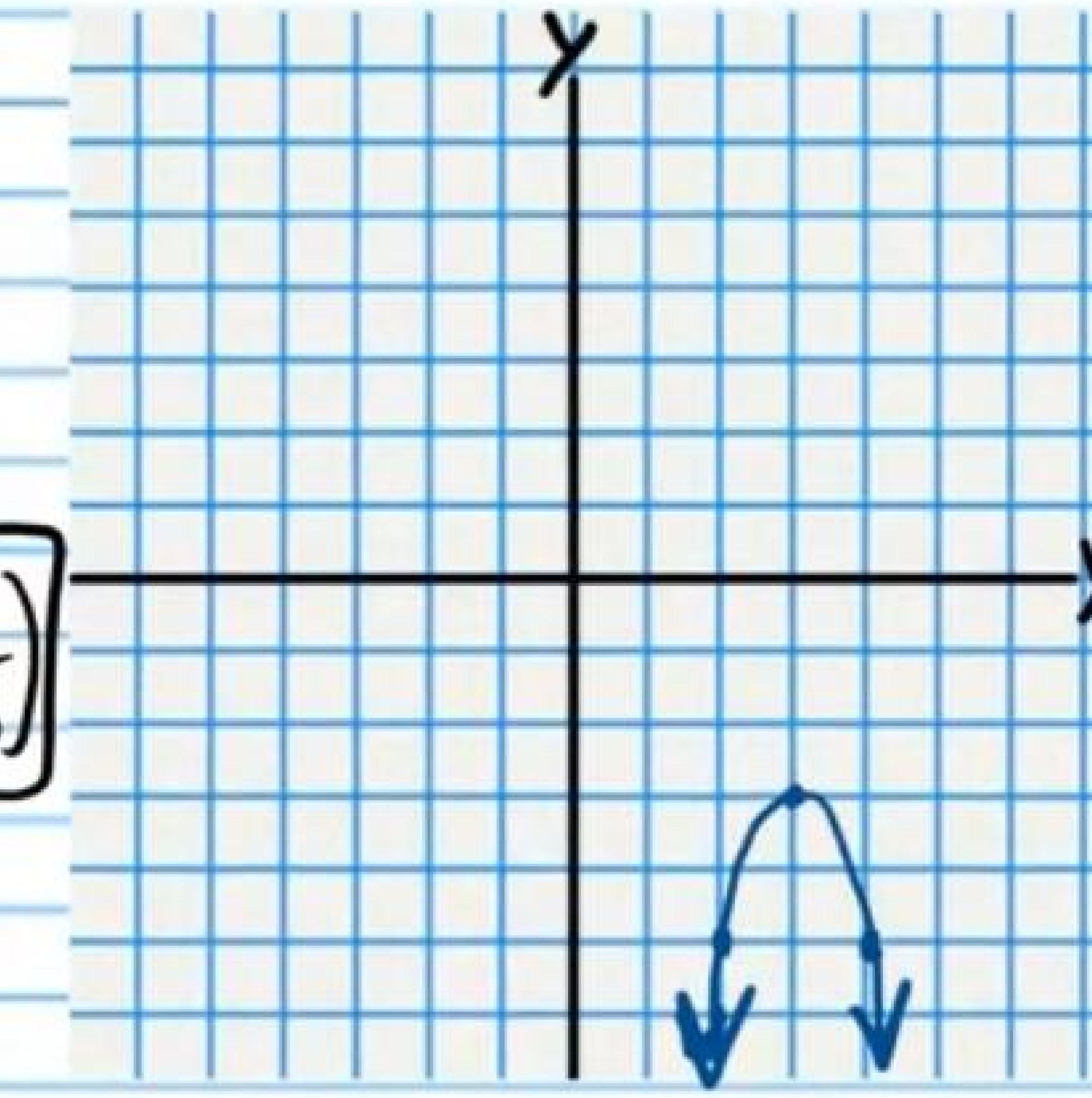
(13) $y+3=-2(x-3)^2$

$$P=\frac{1}{4a}=\frac{1}{4(-2)}=-\frac{1}{8}$$

$$F\left(3,-3-\tfrac{1}{8}\right)=\boxed{F\left(3,-3\tfrac{1}{8}\right)}$$

$$y=-3+\frac{1}{8}$$

$$\boxed{y=-2\frac{7}{8}}$$



(14) $-\frac{1}{4}y^2=x-4$

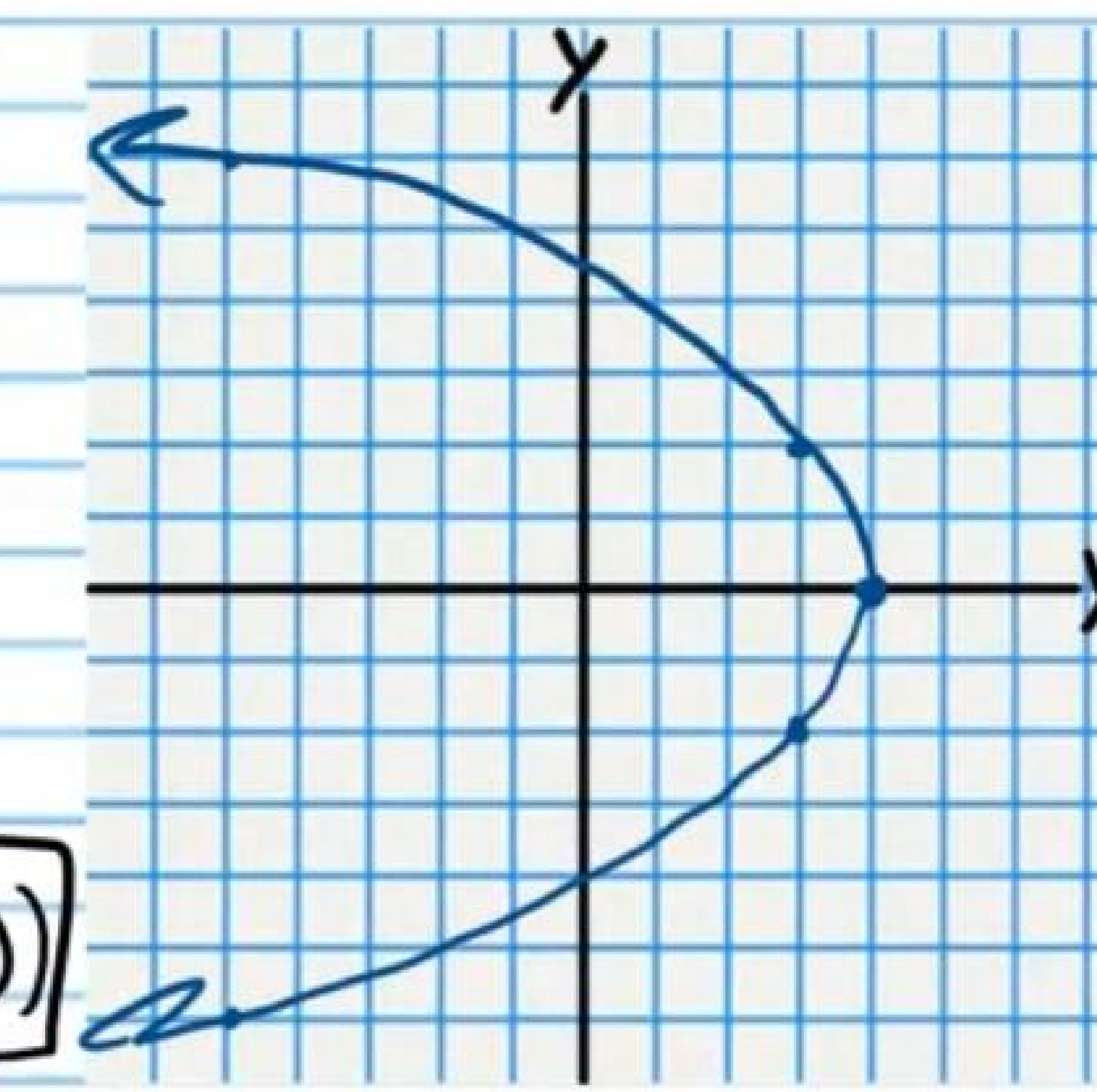
$$-\frac{1}{4}(y-0)^2=x-4$$

$$P=\frac{1}{4a}=\frac{1}{4\left(-\frac{1}{4}\right)}=-1$$

$$F(4-1,0)=\boxed{F(3,0)}$$

$$x=4+1$$

$$\boxed{x=5}$$



$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$ $\frac{(y-k)^2}{a^2}+\frac{(x-h)^2}{b^2}=1$

Ellipses

Graph the following equations. Write the coordinates of the foci.

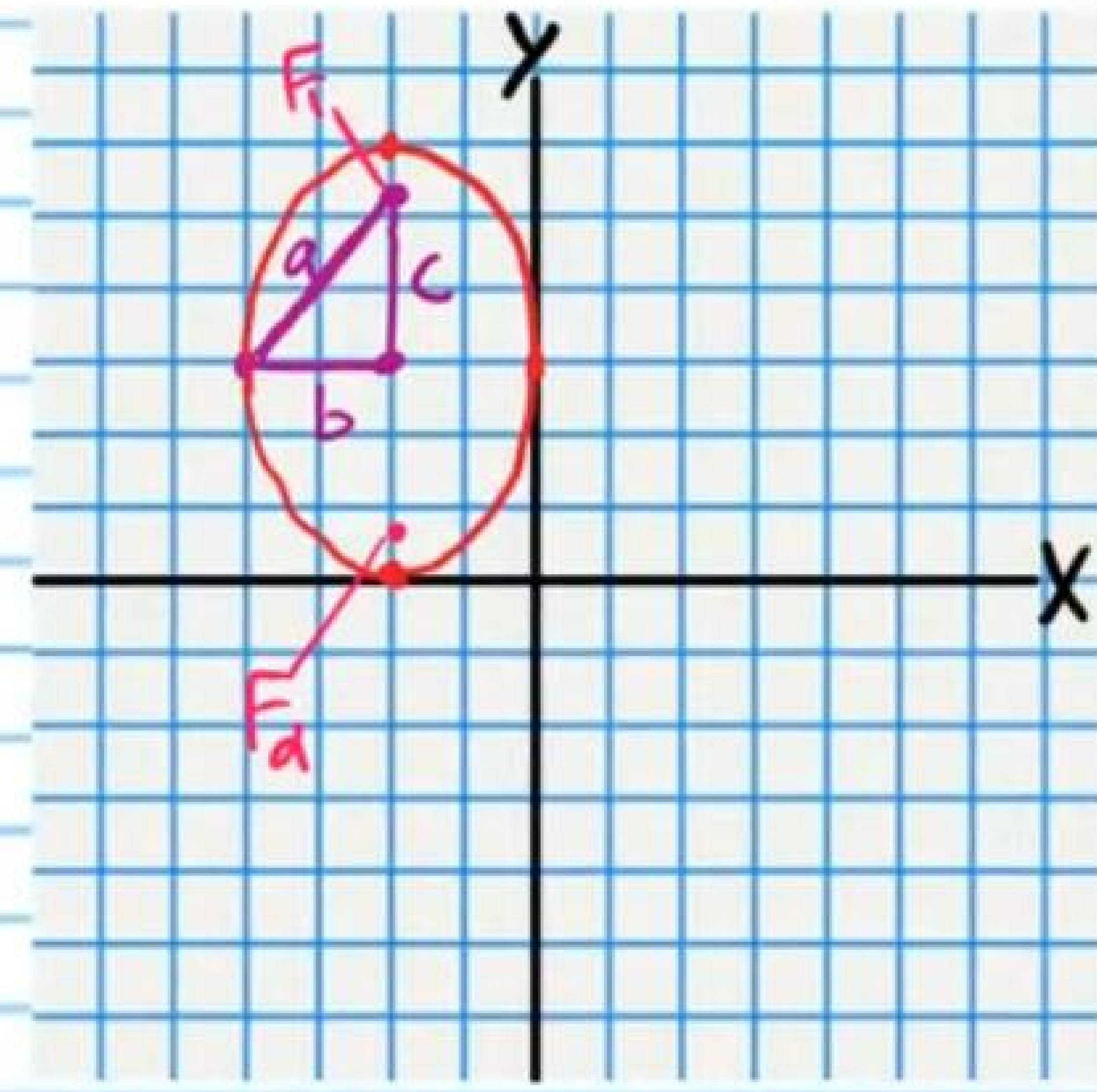
(15) $\frac{(x+2)^2}{4}+\frac{(y-3)^2}{9}=1$

$\frac{(x+2)^2}{2^2}+\frac{(y-3)^2}{3^2}=1$

$\boxed{F_1(-2,3+\sqrt{5})}$ $a^2=b^2+c^2$

$\boxed{F_2(-2,3-\sqrt{5})}$ $3^2=2^2+c^2$

$c=\sqrt{5}$



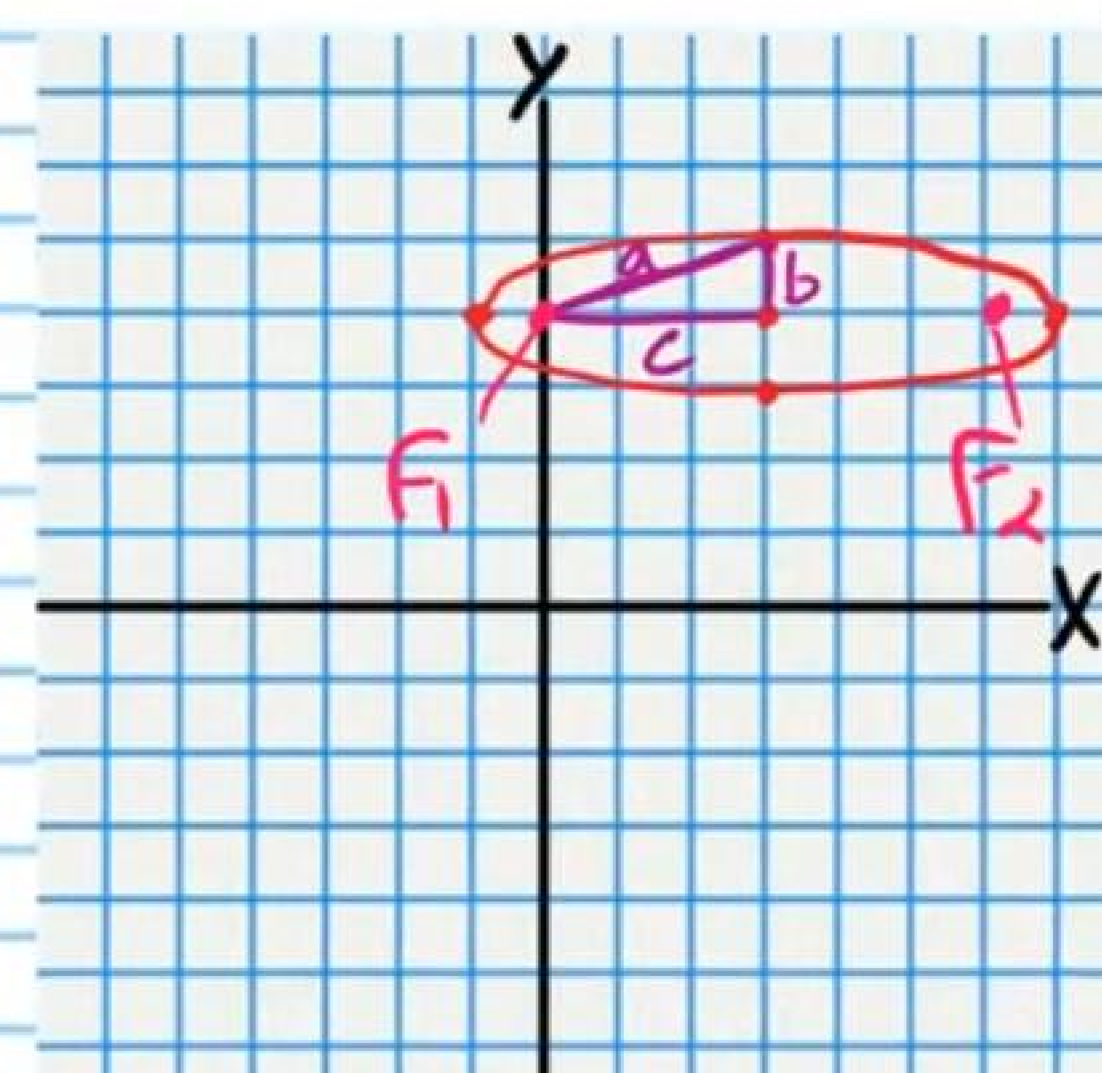
(16) $1=\frac{(x-3)^2}{16}+(y-4)^2$

$1=\frac{(x-3)^2}{4^2}+\frac{(y-4)^2}{1^2}$

$\boxed{\begin{array}{l}F_1(3-\sqrt{15},4)\\F_2(3+\sqrt{15},4)\end{array}}$ $a^2=b^2+c^2$

$4^2=1^2+c$

$c=\sqrt{15}$



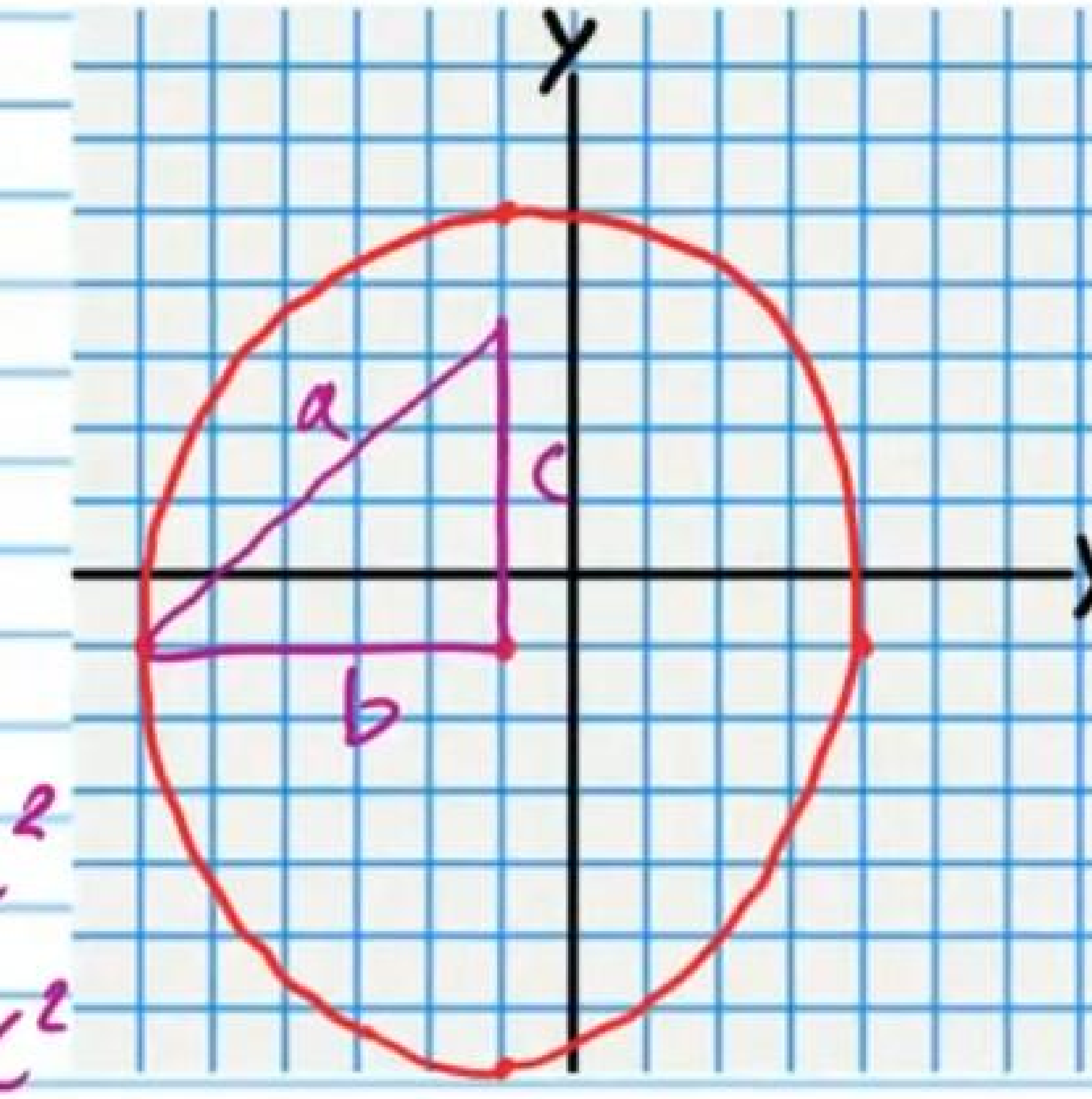
(17) $\frac{(x+1)^2}{25}+\frac{(y+1)^2}{36}=1$

$\frac{(x+1)^2}{5^2}+\frac{(y+1)}{6^2}=1$

$\boxed{\begin{array}{l}F_1(-1,-1+\sqrt{11})\\F_2(-1,-1-\sqrt{11})\end{array}}$ $a^2=b^2+c^2$

$6^2=5^2+c^2$

$c=\sqrt{11}$



(18) $1=\frac{x^2}{16}+\frac{y^2}{7}$

$1=\frac{(x-0)^2}{4^2}+\frac{(y-0)^2}{(\sqrt{7})^2}$

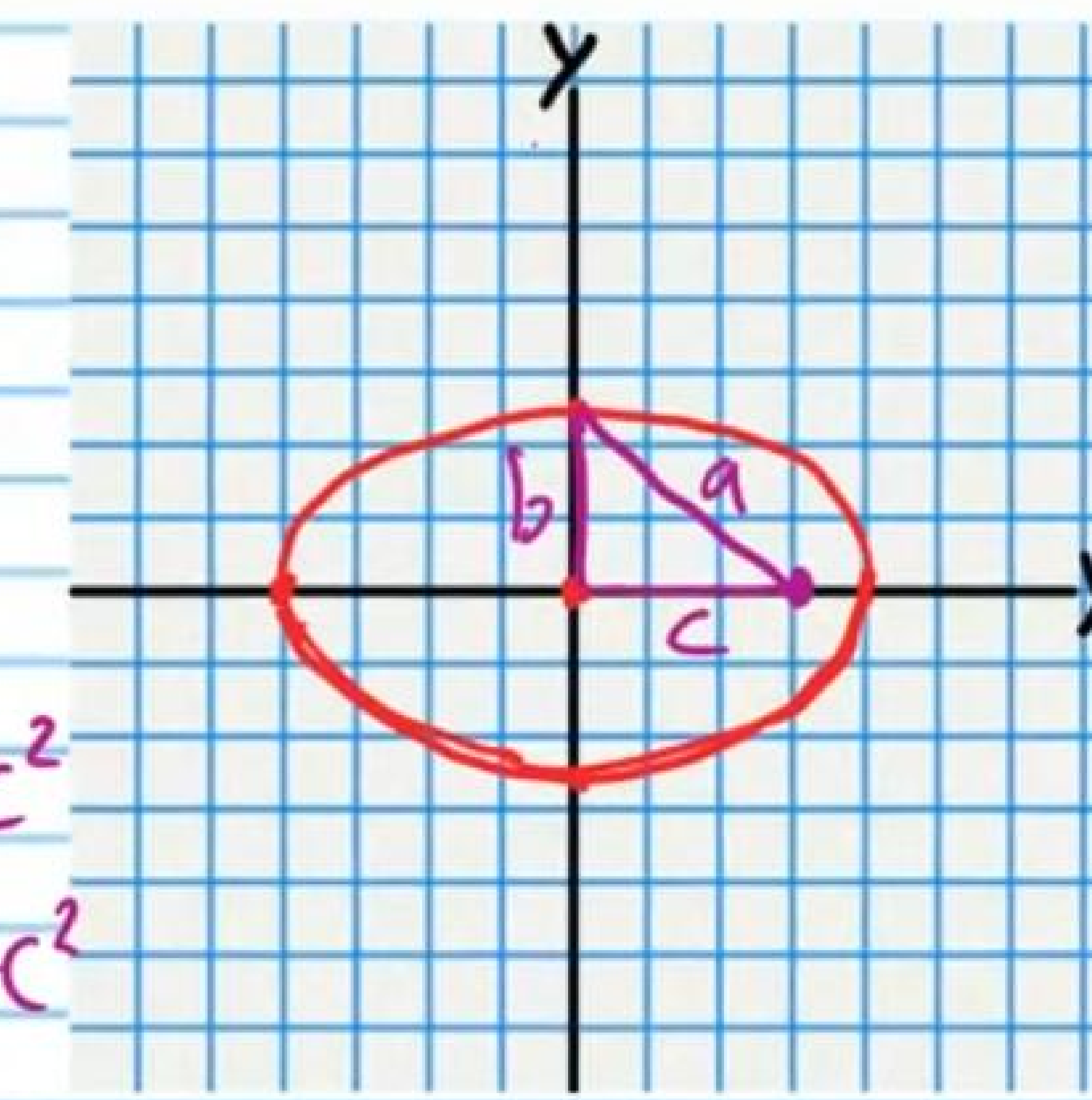
$F_1(0+3,0)$ $a^2=b^2+c^2$

$\boxed{F_1(3,0)}$ $4^2=\sqrt{7}^2+c^2$

$c=3$

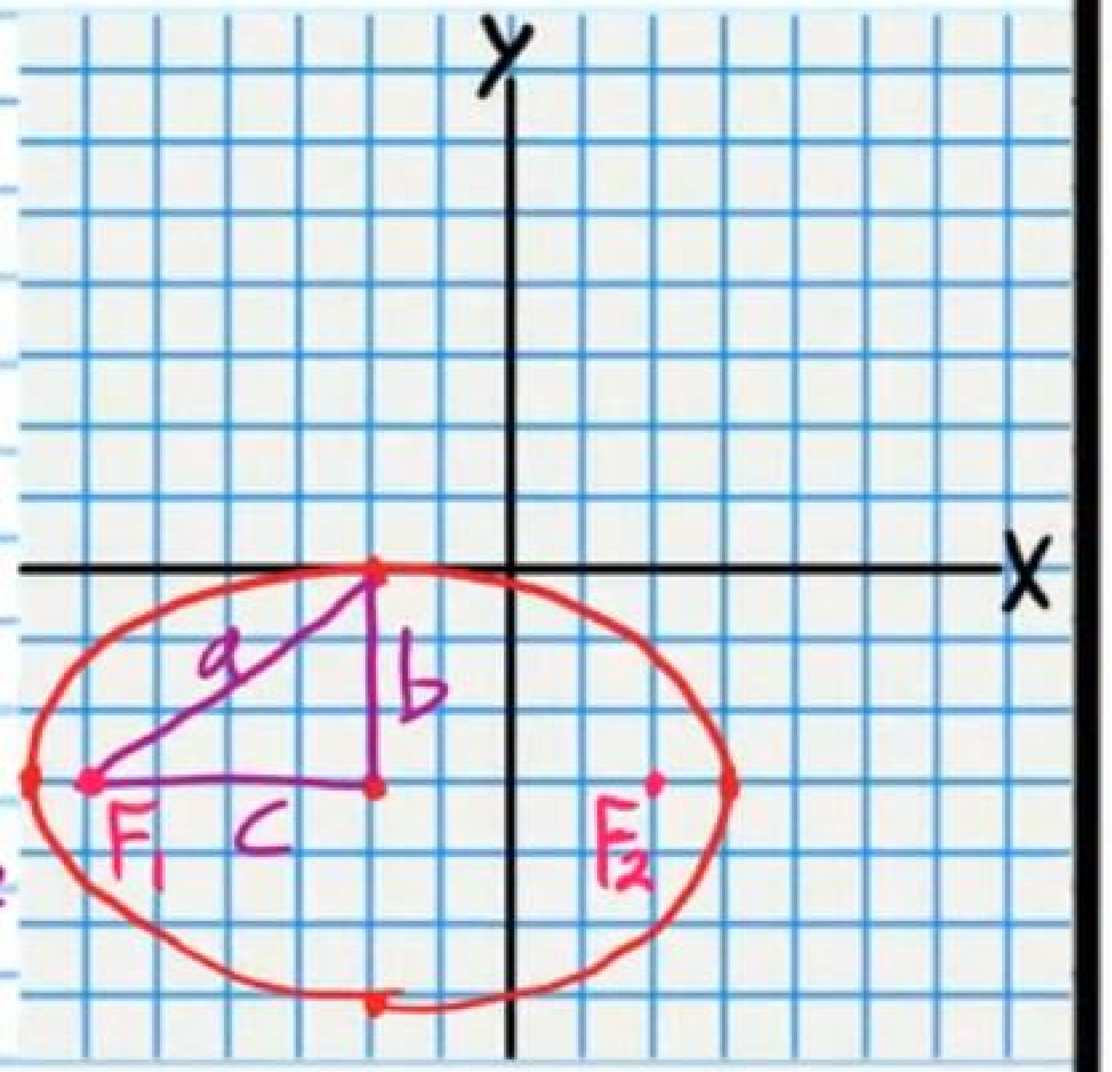
$F_2(0-3,0)$

$\boxed{F_2(-3,0)}$



(19) $\dfrac{(x+2)^2}{25}+\dfrac{(y+3)^2}{9}=1$

$\dfrac{(x+2)^2}{5^2}+\dfrac{(y+3)^2}{3^2}=1$



$F_1(-2-4,-3)$

$\boxed{F_1(-6,-3)}$

$F_2(-2+4,-3)$

$\boxed{F_2(2,-3)}$

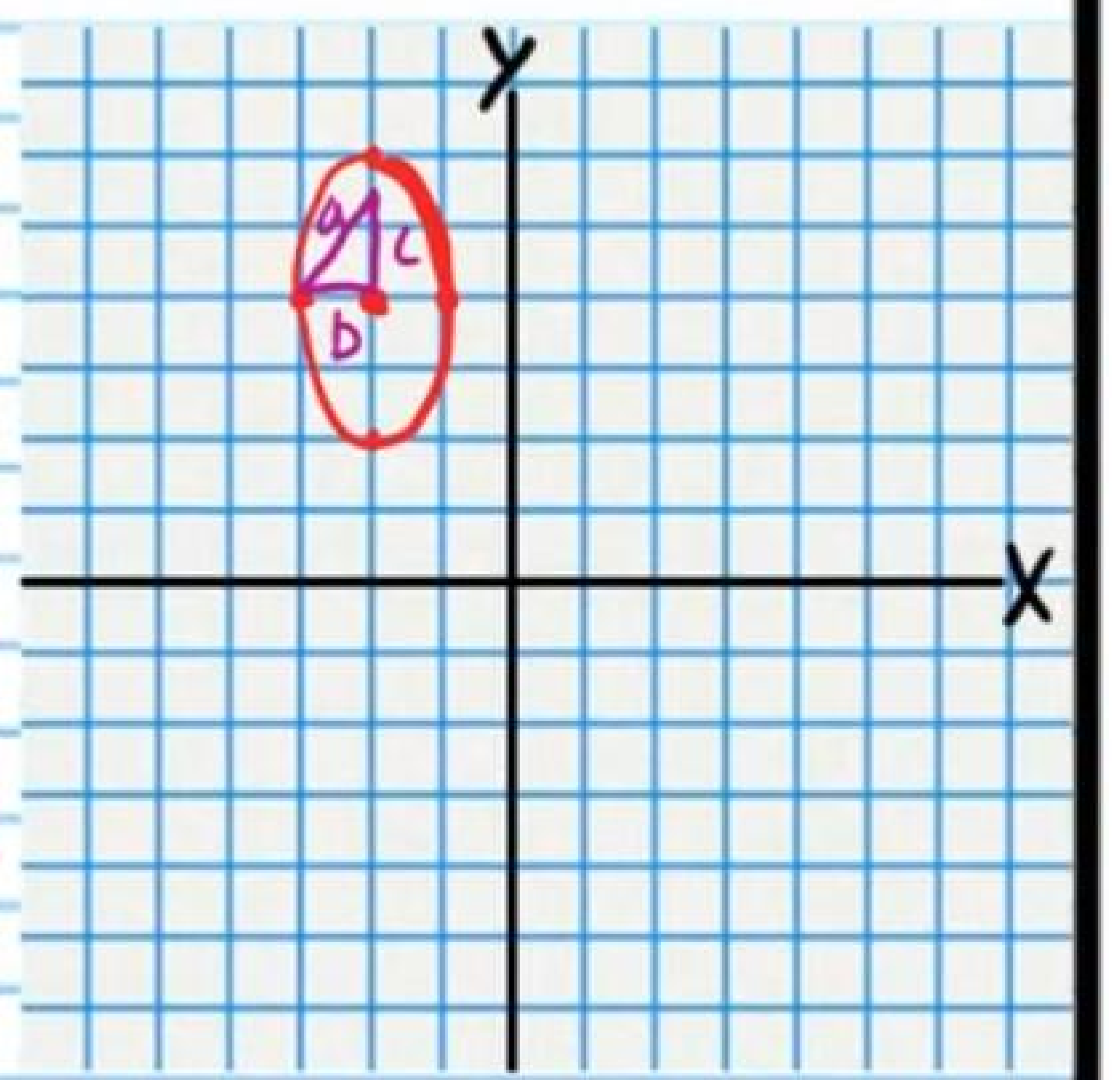
$a^2=b^2+c^2$

$5^2=3^2+c^2$

$c=4$

(20) $1=(x+2)^2+\dfrac{(y-4)^2}{4}$

$1=\dfrac{(x+2)^2}{1^2}+\dfrac{(y-4)^2}{2^2}$



$\boxed{F_1(-2,4+\sqrt{3})}$

$\boxed{F_2(-2,4-\sqrt{3})}$

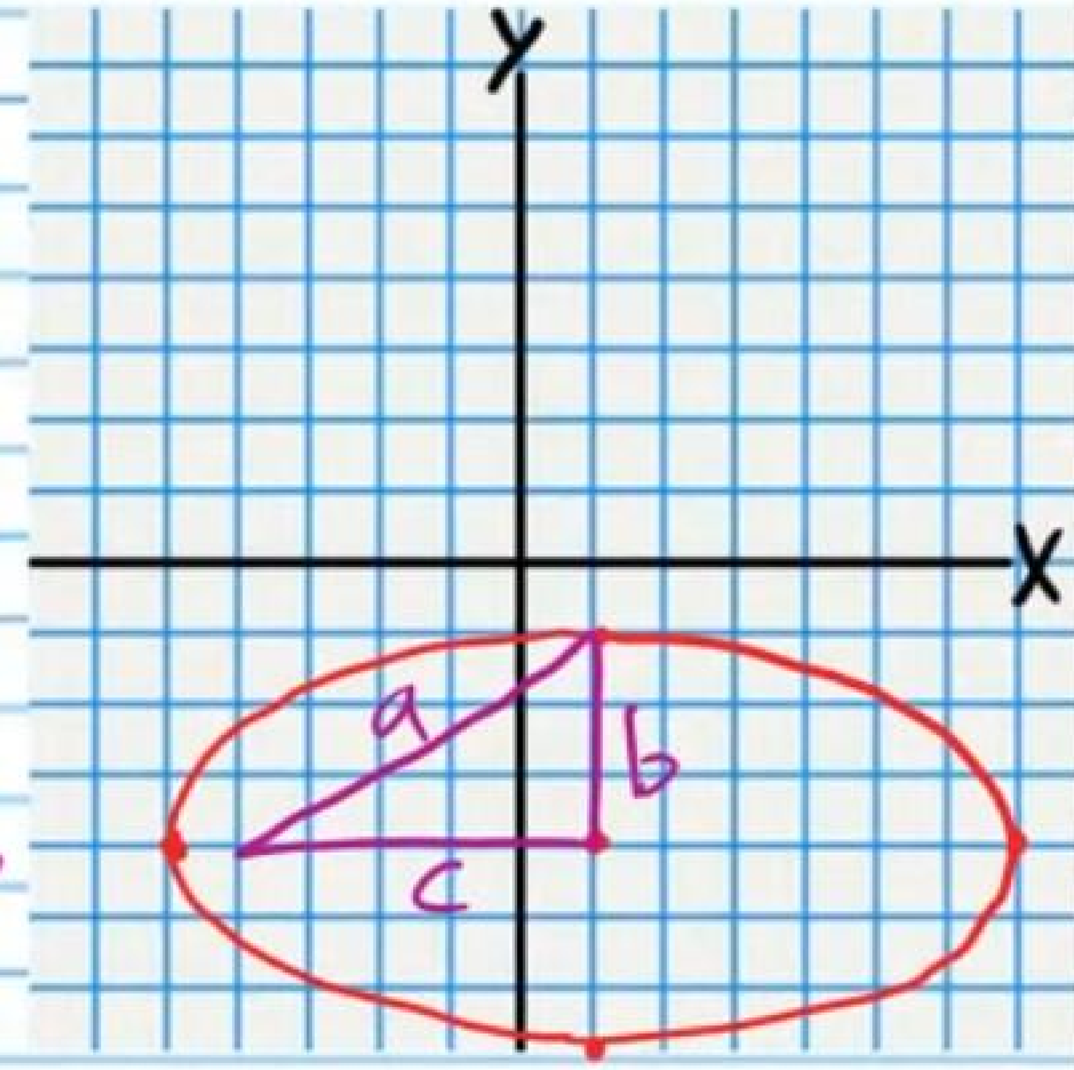
$a^2=b^2+c^2$

$2^2=1^2+c^2$

$c=\sqrt{3}$

(21) $\dfrac{(x-1)^2}{36}+\dfrac{(y+4)^2}{9}=1$

$\dfrac{(x-1)^2}{6^2}+\dfrac{(y+4)^2}{3^2}=1$



$\boxed{F_1(1-3\sqrt{3},-4)}$

$\boxed{F_2(1+3\sqrt{3},-4)}$

$a^2=b^2+c^2$

$6^2=3^2+c^2$

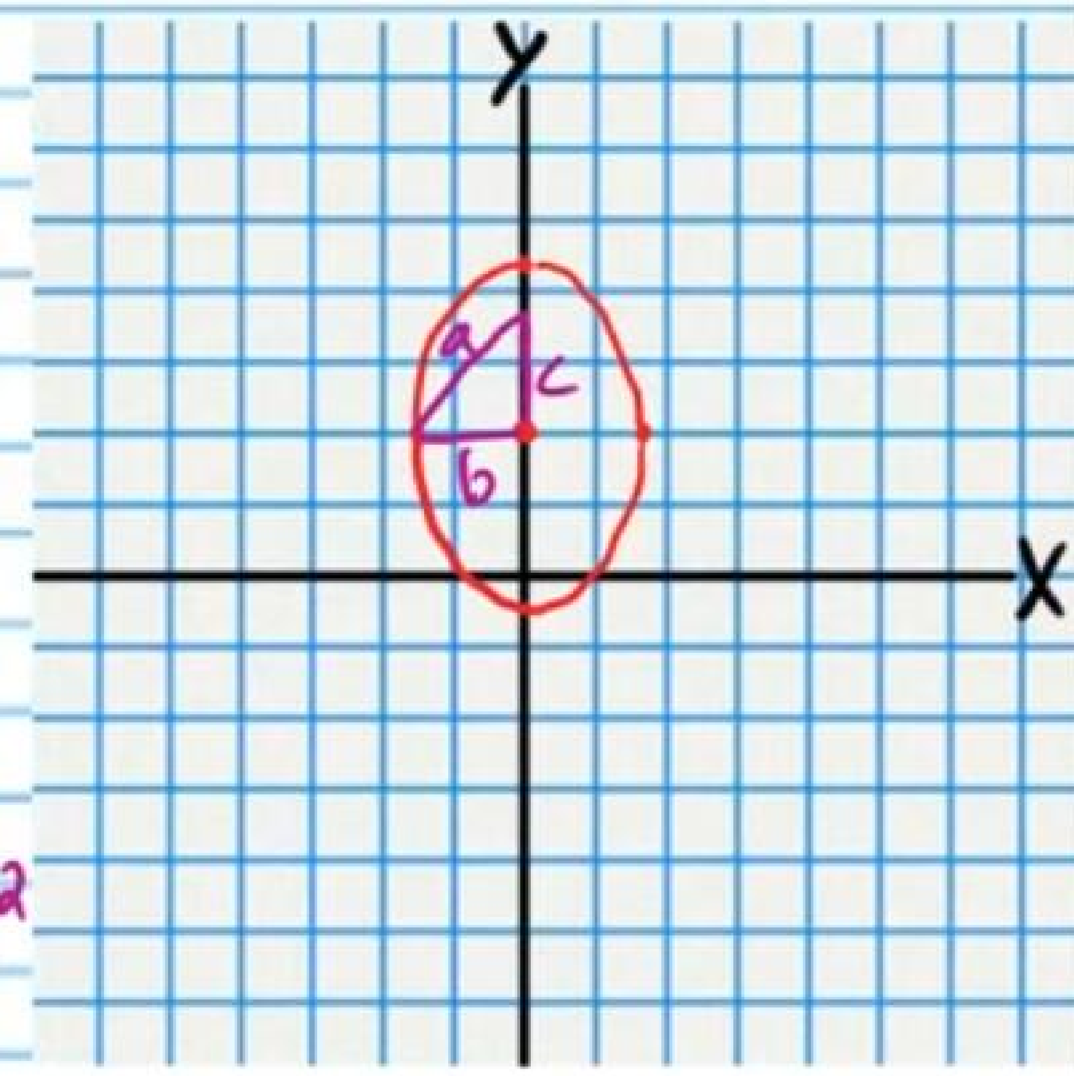
$c=3\sqrt{3}$

$\sqrt{27}=\sqrt{9\cdot 3}=\sqrt{9}\cdot\sqrt{3}$

$=3\sqrt{3}$

(22) $1=\dfrac{x^2}{3}+\dfrac{(y-2)^2}{6}$

$1=\dfrac{(x-0)^2}{(\sqrt{3})^2}+\dfrac{(y-2)^2}{(\sqrt{6})^2}$



$\boxed{F_1(0,2+\sqrt{3})}$

$\boxed{F_2(0,2-\sqrt{3})}$

$a^2=b^2+c^2$

$(\sqrt{6})^2=(\sqrt{3})^2+c^2$

$c=\sqrt{3}$

$\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$

$x-h=\frac{b}{a}(y-k)$

$x-h=-\frac{b}{a}(y-k)$

Hyperbolas

$\frac{(y-k)^2}{a^2}-\frac{(x-h)^2}{b^2}=1$

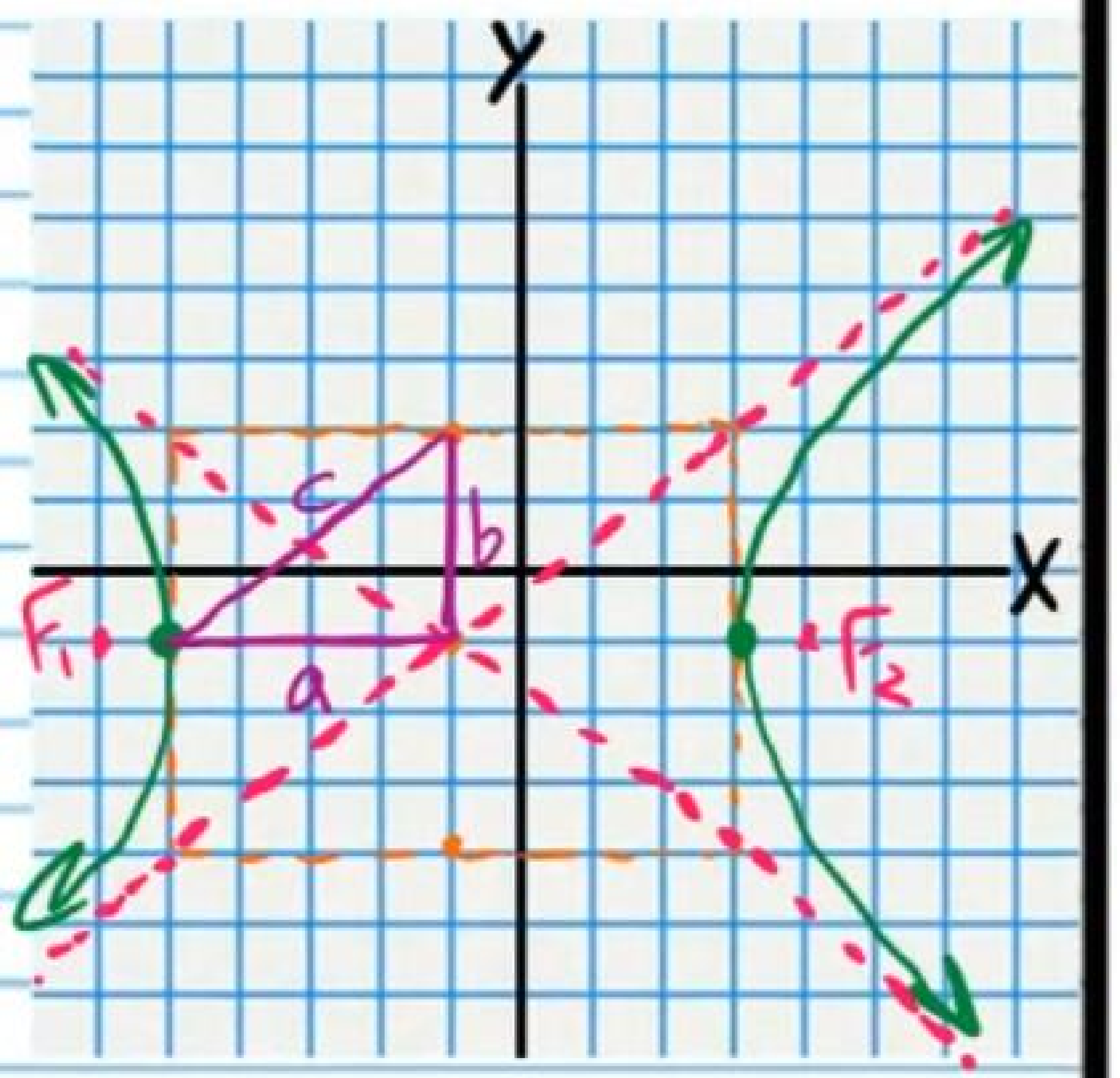
$y-k=\frac{a}{b}(x-h)$

$y-k=-\frac{a}{b}(x-h)$

Graph the following equations. Write the coordinates of the foci, draw the asymptotes as dotted lines, and write their equations.

(23) $\frac{(x+1)^2}{16}-\frac{(y+1)^2}{9}=1$

$\frac{(x+1)^2}{4^2}-\frac{(y+1)}{3^2}=1$



$F_1(-1-5,-1)$

$\boxed{F_1(-6,-1)}$

$a^2+b^2=c^2$

$4^2+3^2=c^2$

$c=5$

$F_2(-1+5,-1)$

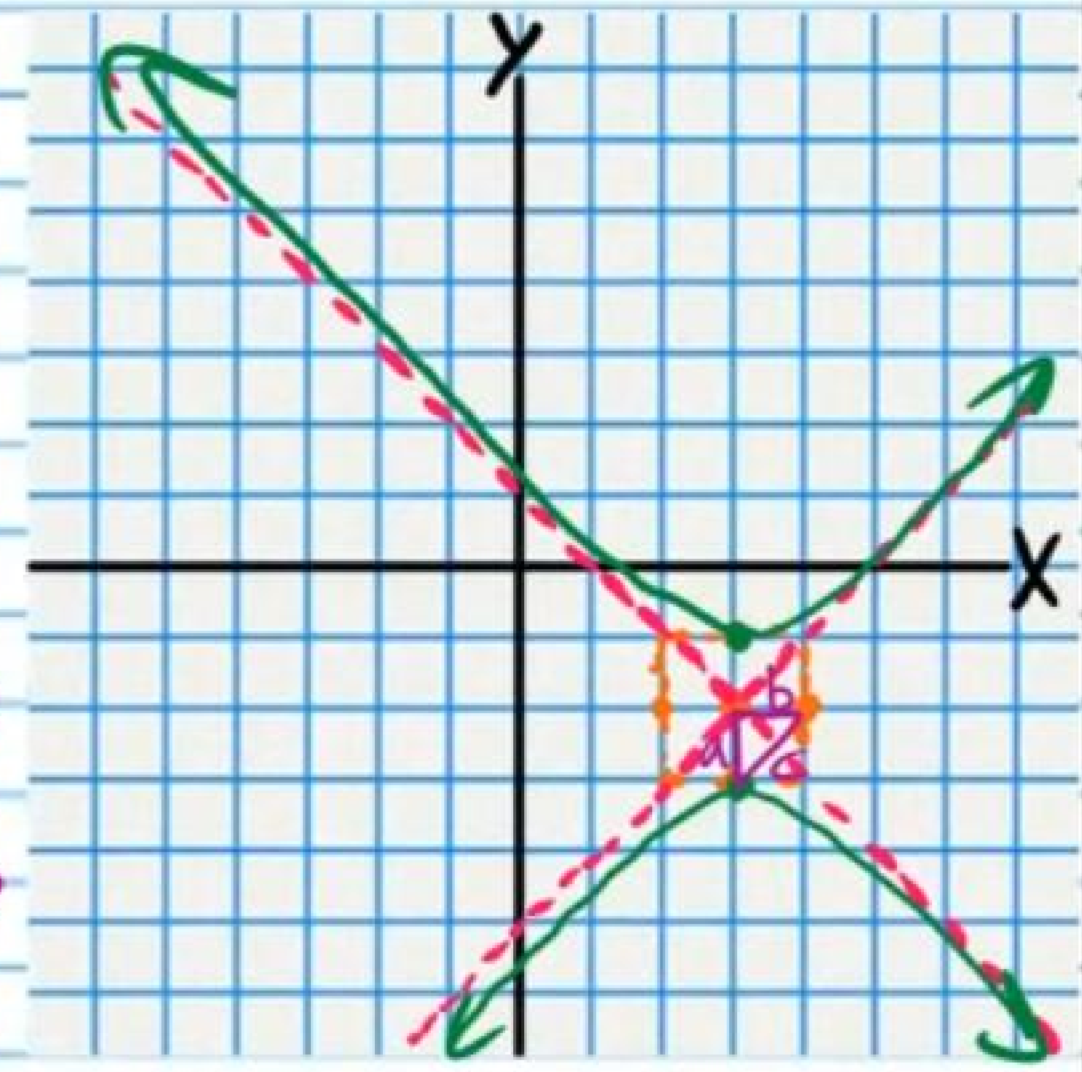
$\boxed{F_2(4,-1)}$

Asymptotes

$\boxed{y+1=\frac{3}{4}(x+1) \quad y+1=-\frac{3}{4}(x+1)}$

(24) $1=(y+2)^2-(x-3)^2$

$1=\frac{(y+2)^2}{1^2}-\frac{(x-3)^2}{1^2}$



$\boxed{F_1(3,-2-\sqrt{2})}$

$\boxed{F_2(3,-2+\sqrt{2})}$

$a^2+b^2=c^2$

$1^2+1^2=c^2$

$c=\sqrt{2}$

Asymptotes

$P(3,-2) \quad m_1=1$

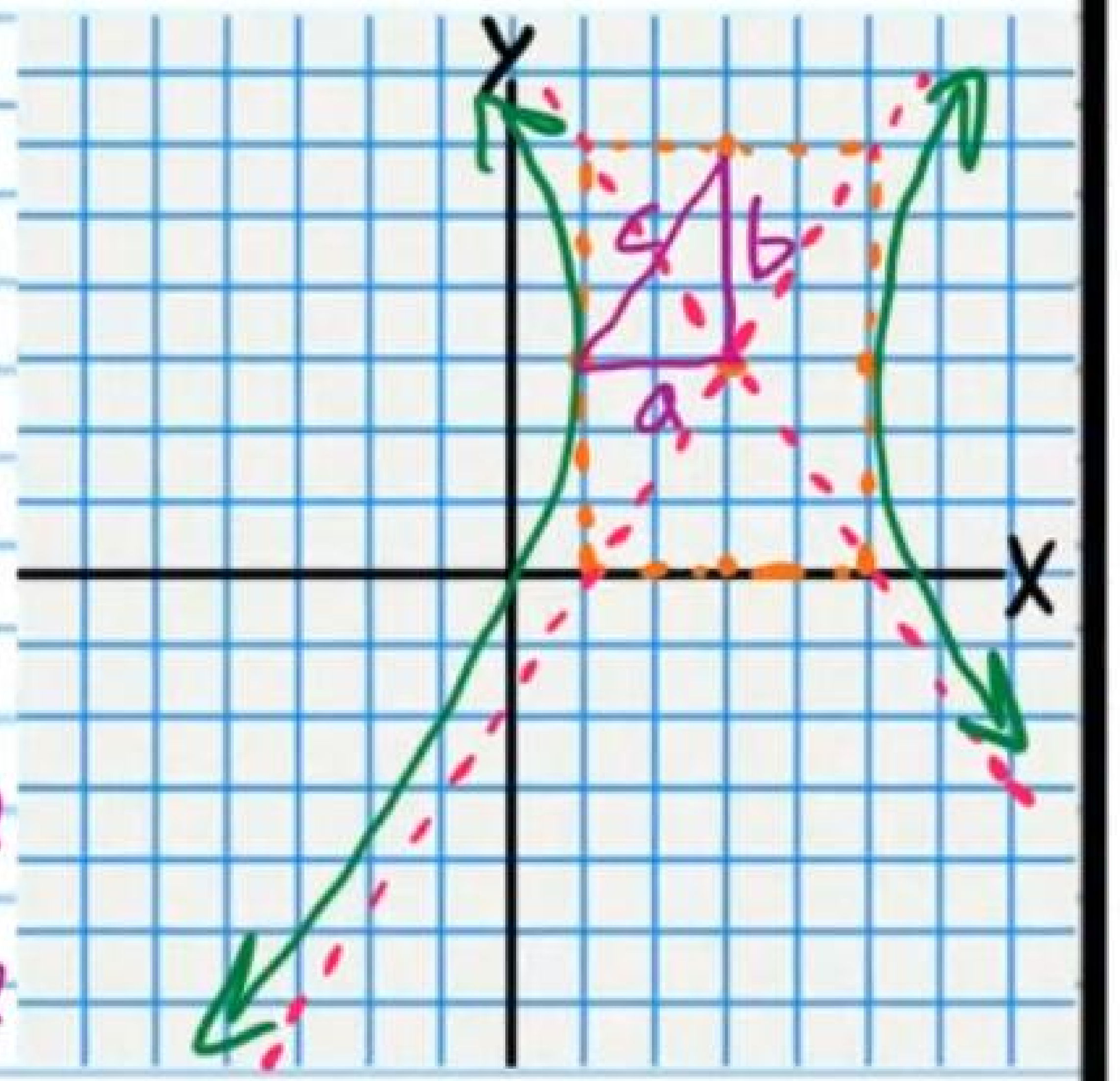
$m_2=-1$

$y+2=1(x-3)$

$\boxed{y+2=-(x-3)}$

$\boxed{y+2=x-3}$

25) $\frac{(x-3)^2}{4} - \frac{(y-3)^2}{9} = 1$



$\frac{(x-3)^2}{2^2} - \frac{(y-3)^2}{3^2} = 1$

$\boxed{F_1(3-\sqrt{13}, 3)}$

$\boxed{F_2(3+\sqrt{13}, 3)}$

$a^2+b^2=c^2$

$2^2+3^2=c^2$

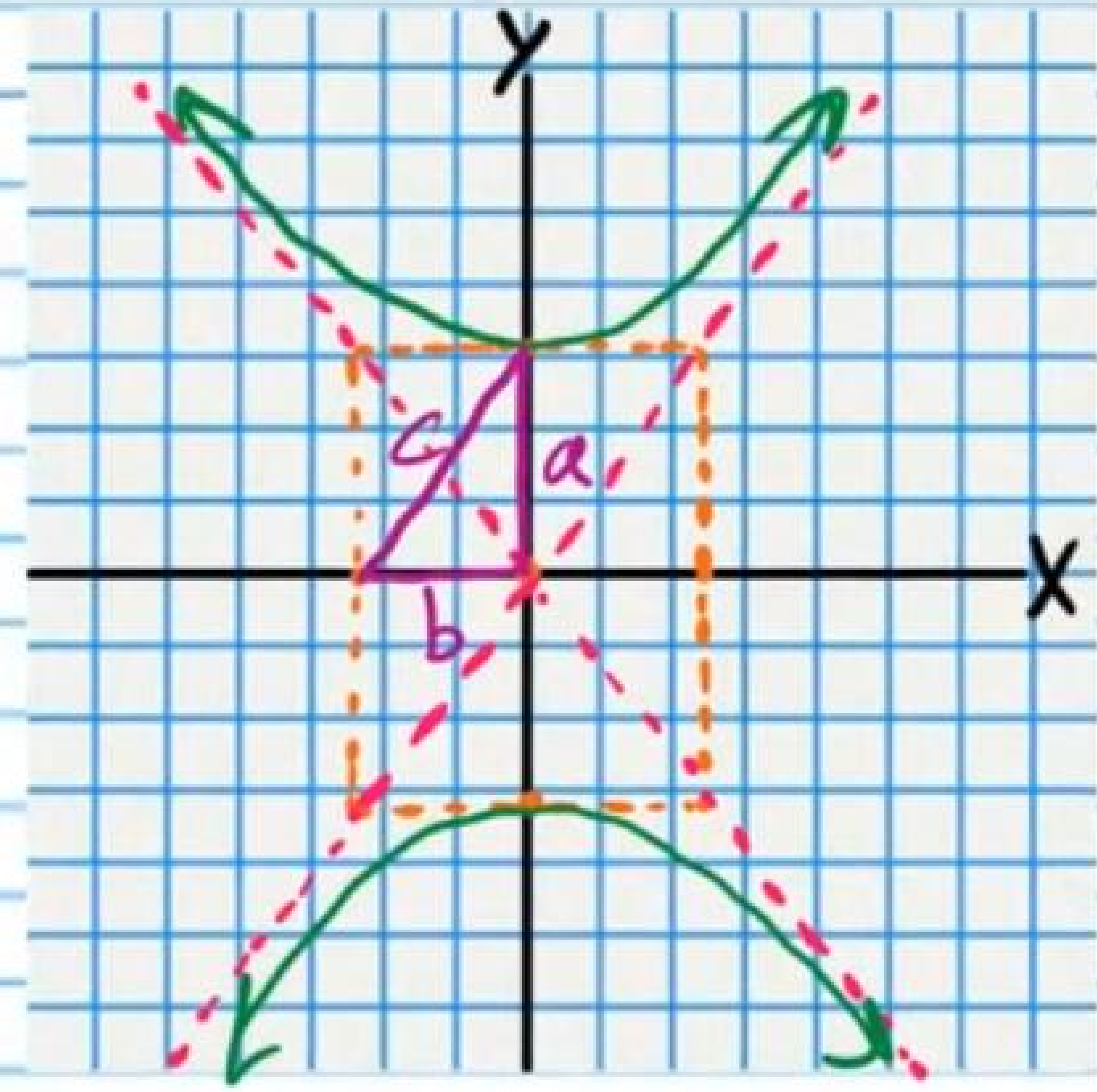
$c=\sqrt{13}$

Asymptotes

$\boxed{y-3=\frac{3}{2}(x-3)}$

$\boxed{y-3=-\frac{3}{2}(x-3)}$

26) $1=\frac{y^2}{10} - \frac{x^2}{6}$



$1=\frac{(y-0)^2}{(\sqrt{10})^2} - \frac{(x-0)^2}{(\sqrt{6})^2}$

$a^2+b^2=c^2$

$(\sqrt{10})^2+(\sqrt{6})^2=c^2$

$c=4$

$F_1(0,0+4)$ $F_2(0,0-4)$

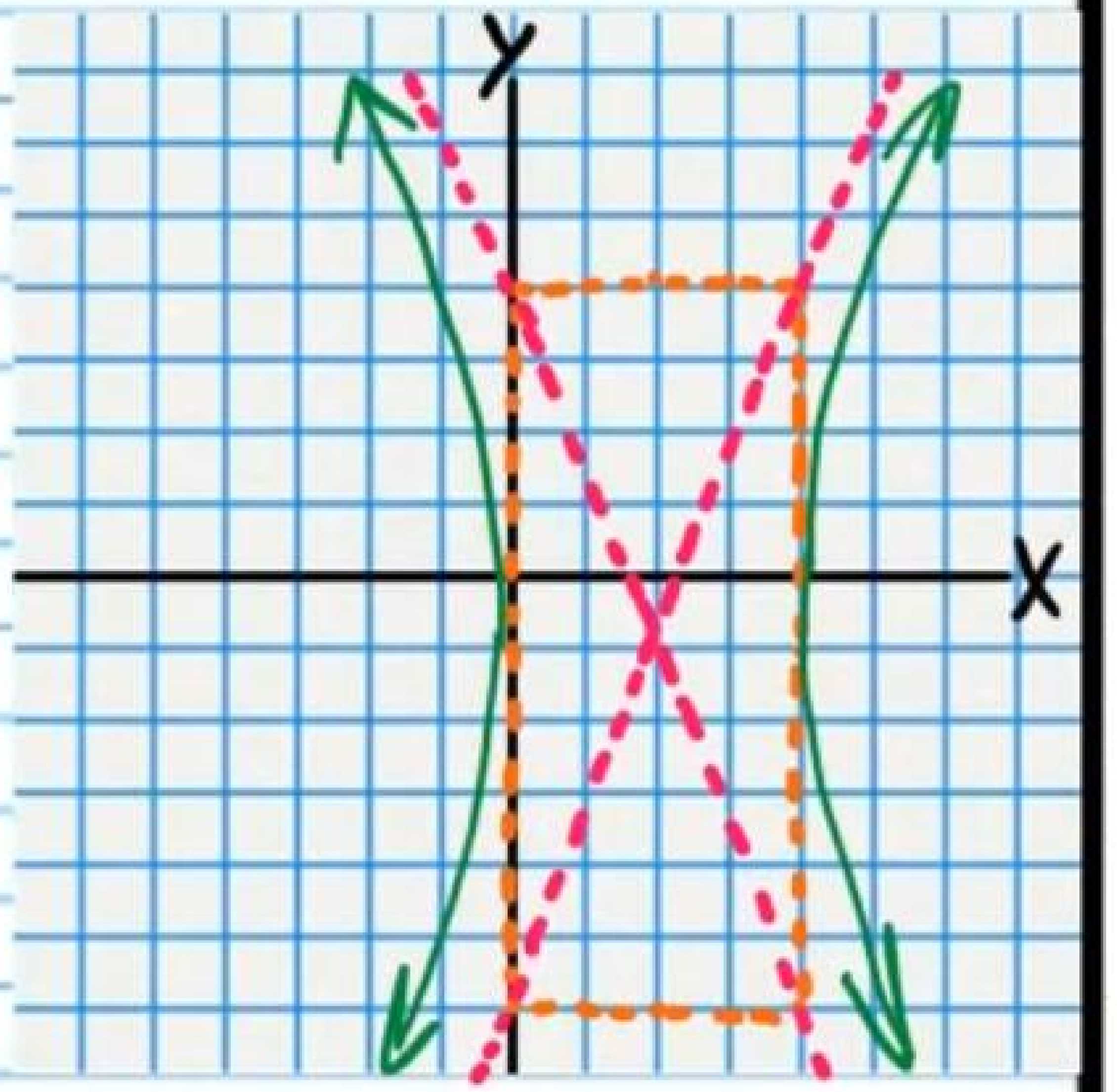
Asymptotes

$\boxed{F_1(0,4)}$ $\boxed{F_2(0,-4)}$

$y-0=\frac{\sqrt{10}}{\sqrt{6}}(x-0)$ $y-0=-\frac{\sqrt{10}}{\sqrt{6}}(x-0)$

$\boxed{y=\frac{\sqrt{60}}{6}x}$ $\boxed{y=-\frac{\sqrt{60}}{6}x}$

(27) $1 = \frac{(x-2)^2}{4} - \frac{(y+1)^2}{25}$

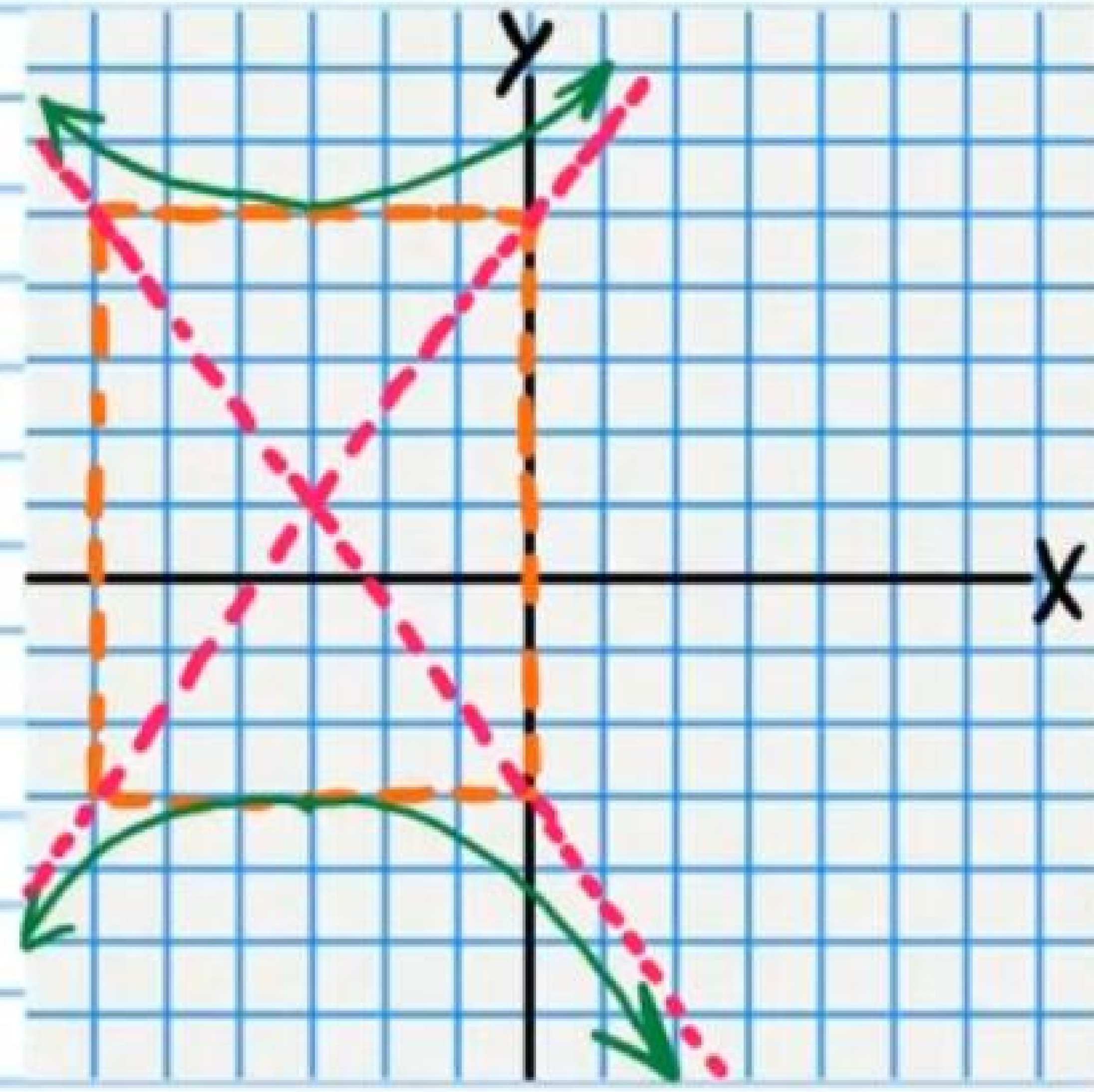


$\boxed{F_1(2-\sqrt{29},-1)}$ $\boxed{F_2(2+\sqrt{29},-1)}$

Asymptotes

$\boxed{y+1 = \frac{5}{2}(x-2)}$ $\boxed{y+1 = -\frac{5}{2}(x-2)}$

(28) $\frac{(y-1)^2}{16} - \frac{(x+3)^2}{9} = 1$

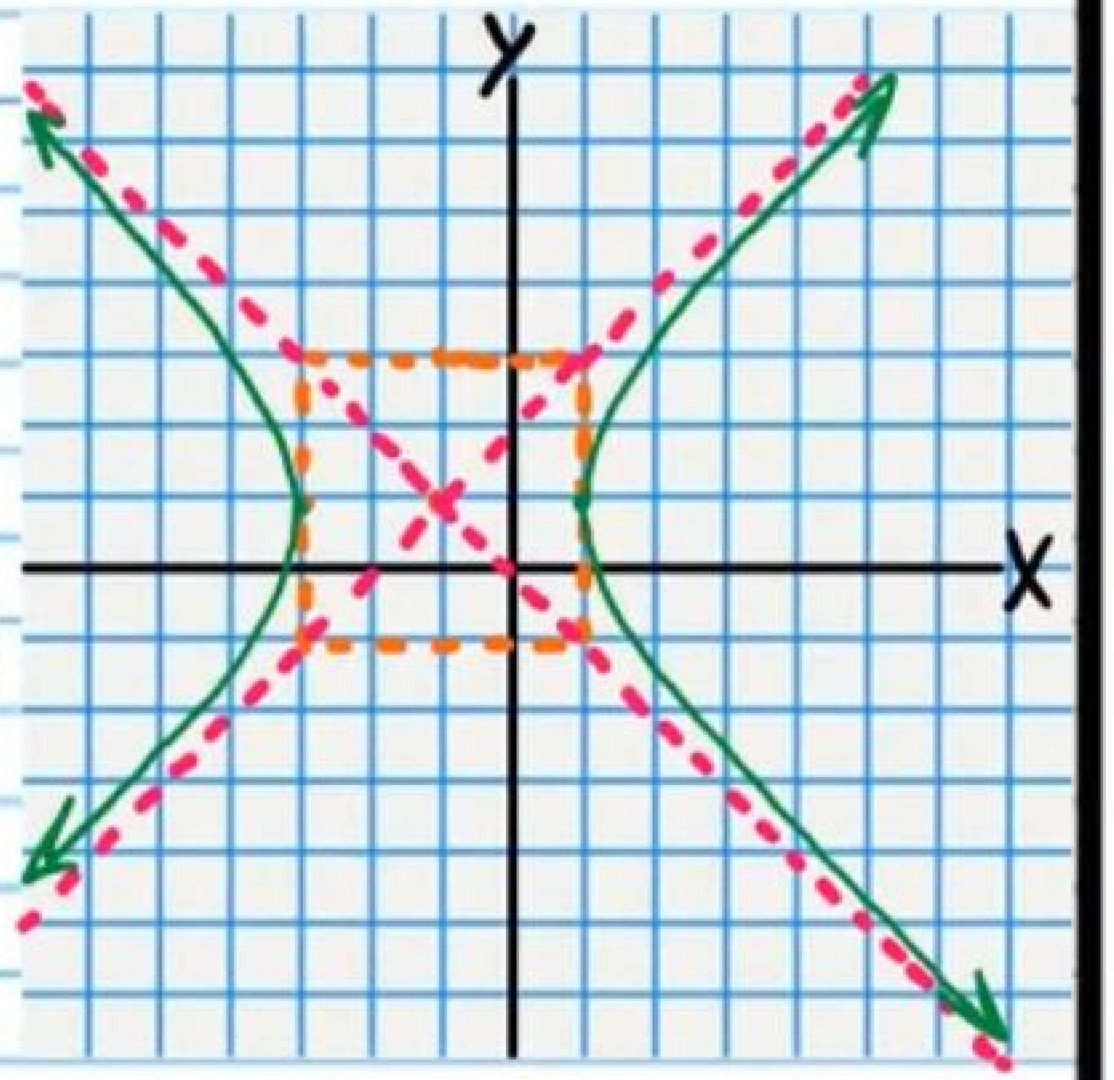


$\boxed{F_1(-3,6)}$ $\boxed{F_2(-3,-4)}$

Asymptotes

$\boxed{y-1 = \frac{4}{3}(x+3)}$ $\boxed{y-1 = -\frac{4}{3}(x+3)}$

(29) $1 = \frac{(x+1)^2}{4} - \frac{(y-1)^2}{4}$

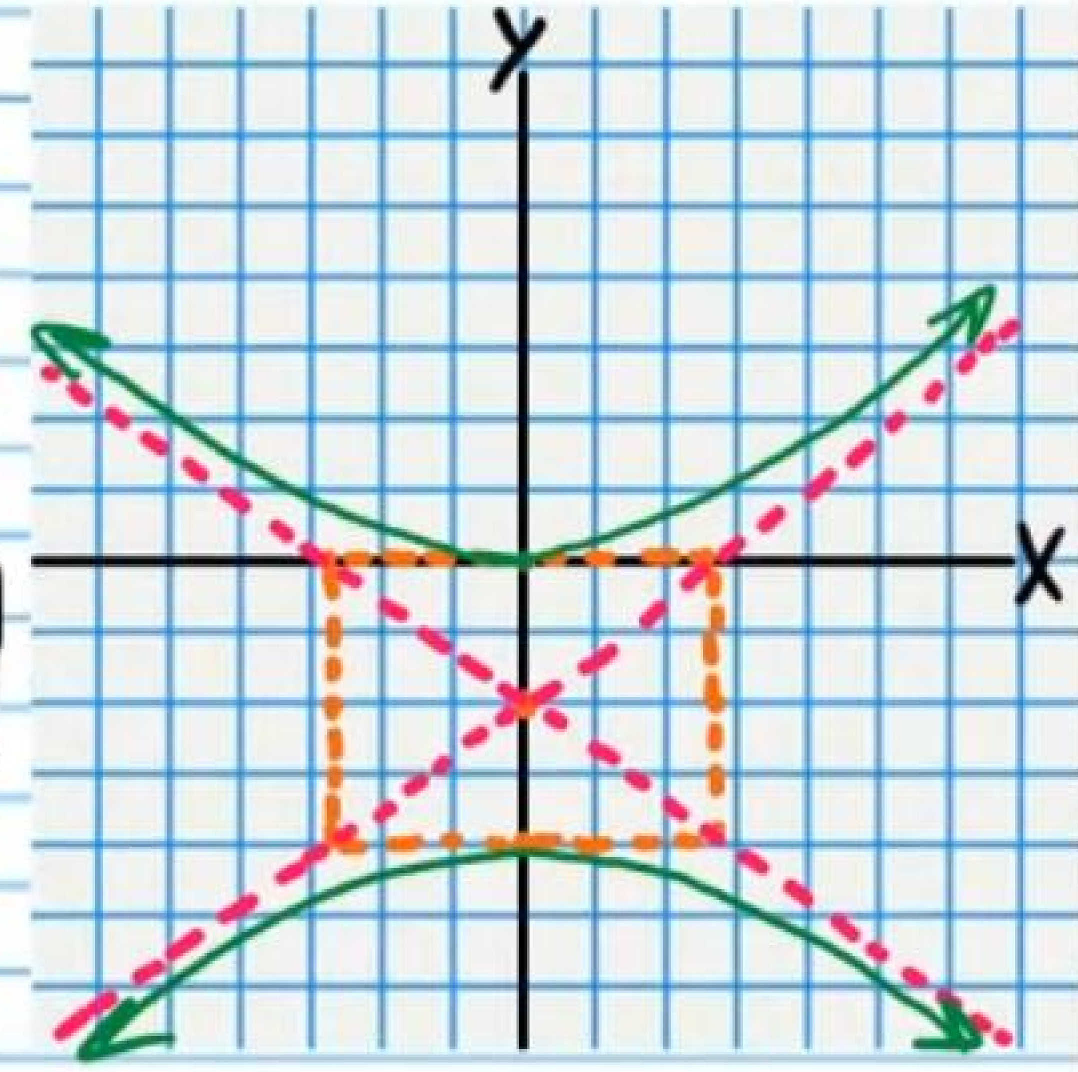


$\boxed{F_1(-1+2\sqrt{2}, 1)}$ $\boxed{F_2(-1-2\sqrt{2}, 1)}$

Asymptotes

$\boxed{y-1 = x+1}$ $\boxed{y-1 = -(x+1)}$

(30) $\frac{(y+2)^2}{4} - \frac{x^2}{8} = 1$



$\boxed{F_1(0, -2+2\sqrt{3})}$ $\boxed{F_2(0, -2-2\sqrt{3})}$

Asymptotes

$y+2 = \frac{\sqrt{2}}{2}(x-0)$ $y+2 = -\frac{\sqrt{2}}{2}(x-0)$

$\boxed{y+2 = \frac{\sqrt{2}}{2}x}$ $\boxed{y+2 = -\frac{\sqrt{2}}{2}x}$